

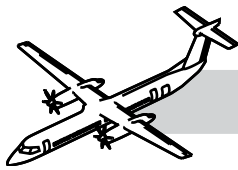
CHAPTER 71 POWERPLANT

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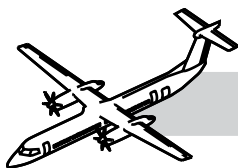
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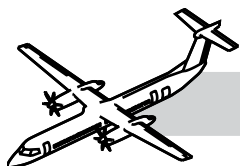
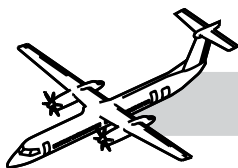
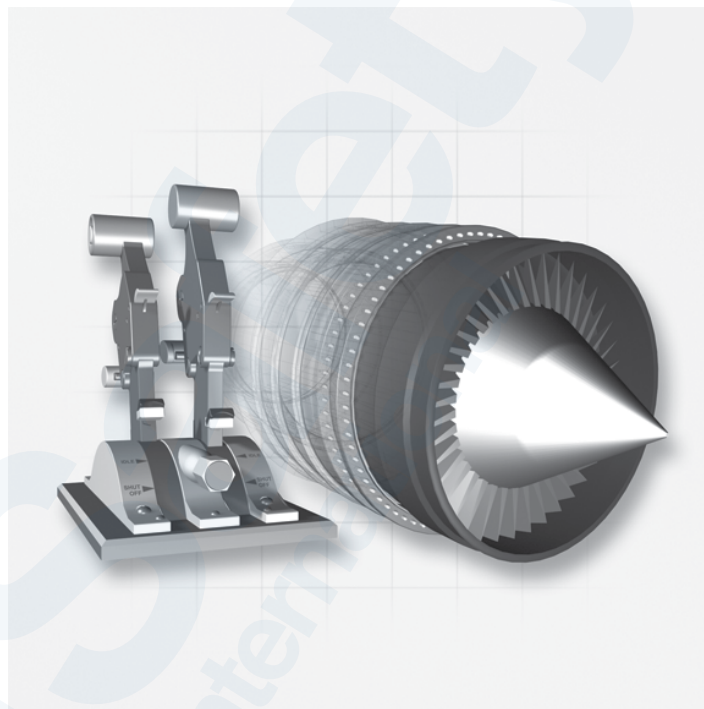


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CHAPTER 71 POWERPLANT



71 POWERPLANT

71-00-00 INTRODUCTION

The PW150A engine provides motive power to the aircraft and also supplies power to various aircraft systems.

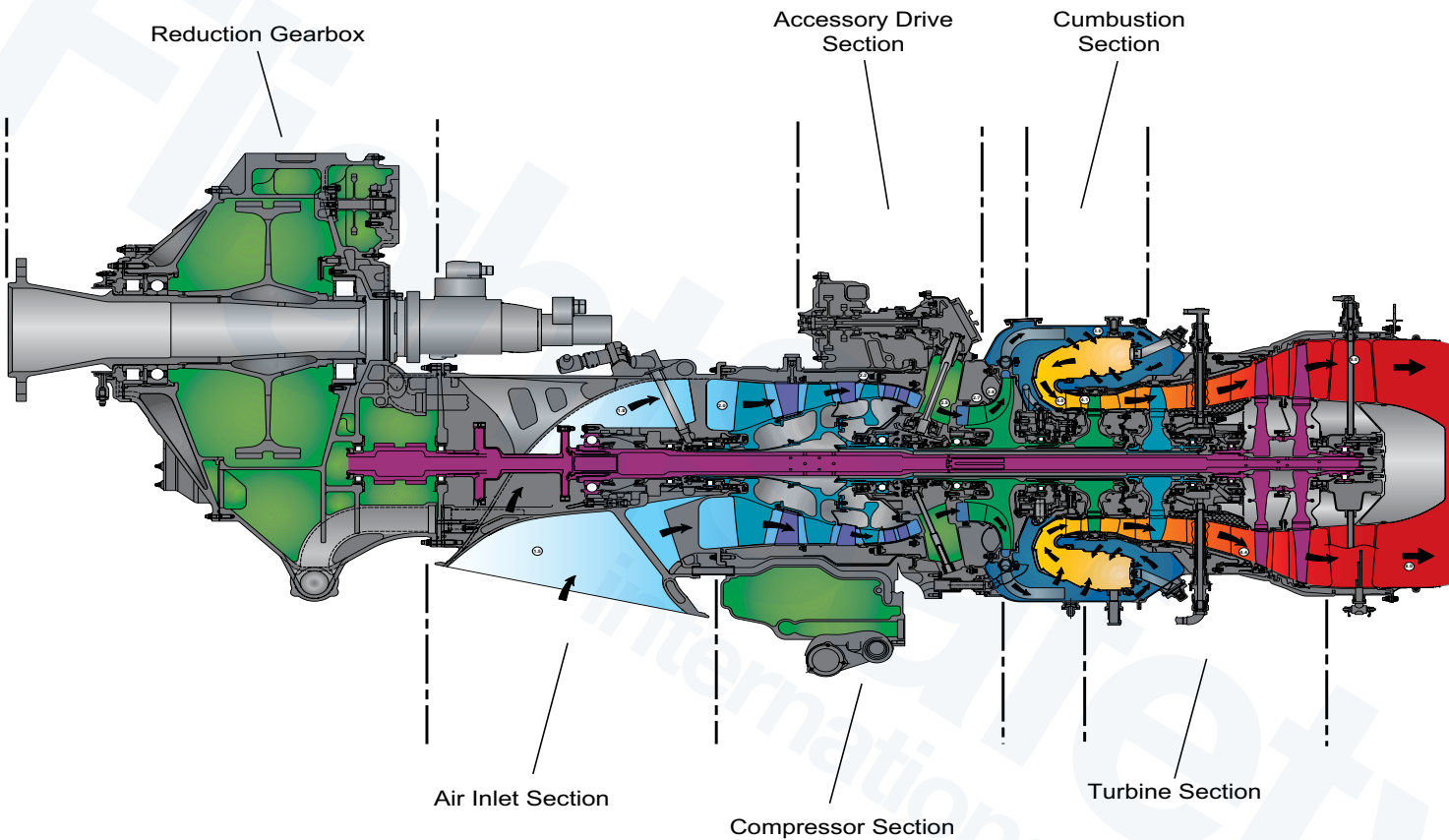
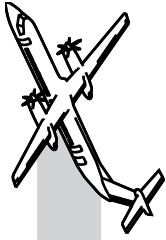
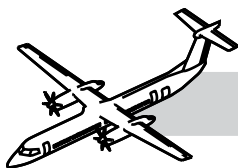


Figure 71-1. PW150A Engine Cross Section

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GENERAL

NOTES

Refer to Figure 71-1. PW150A Engine Cross Section.

The engine has two modules, the reduction gearbox module and the turbo-machinery module.

This section will explain the maintenance procedures for the removal and installation of the following items:

- The engine from the nacelle
- The nacelle mounted engine hoist

This section will also include description of the following components:

- Cowlings
- Engine isolator system
- Hydraulic torque compensation system
- Forward nacelle intakes and
- Engine nacelle drains.

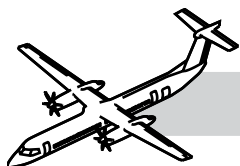
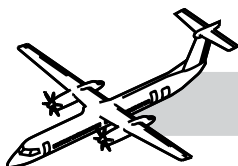


Figure 71-2. Propeller Removal



REMOVAL AND INSTALLATION

NOTES

Engine Hoist removal and Installation

Refer to:

- Figure 71-2. Propeller Removal.
- Figure 71-3. Engine Hoist.

The following procedure is used for the installation of the nacelle mounted engine hoist.

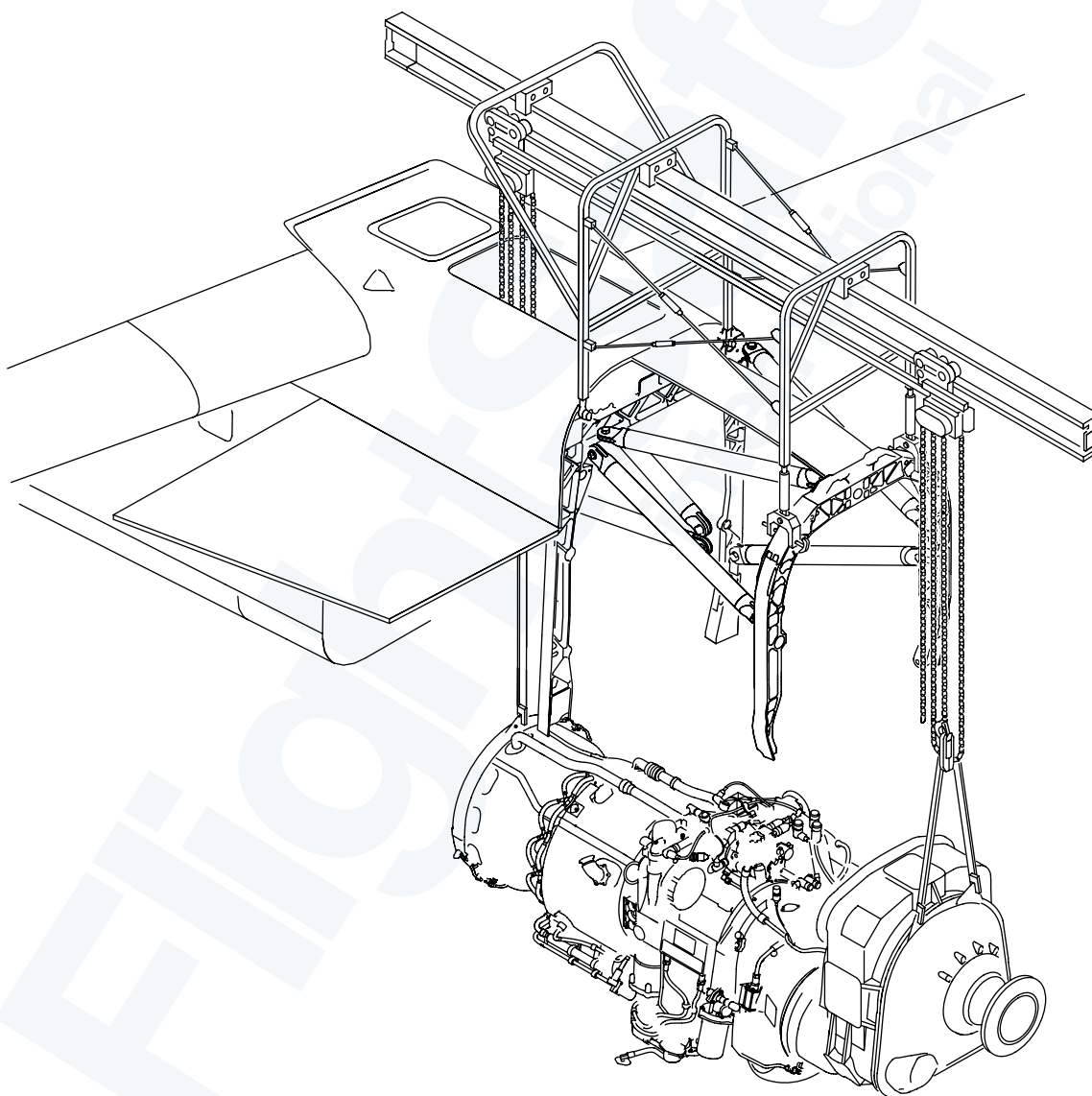


Figure 71-3. Engine Hoist

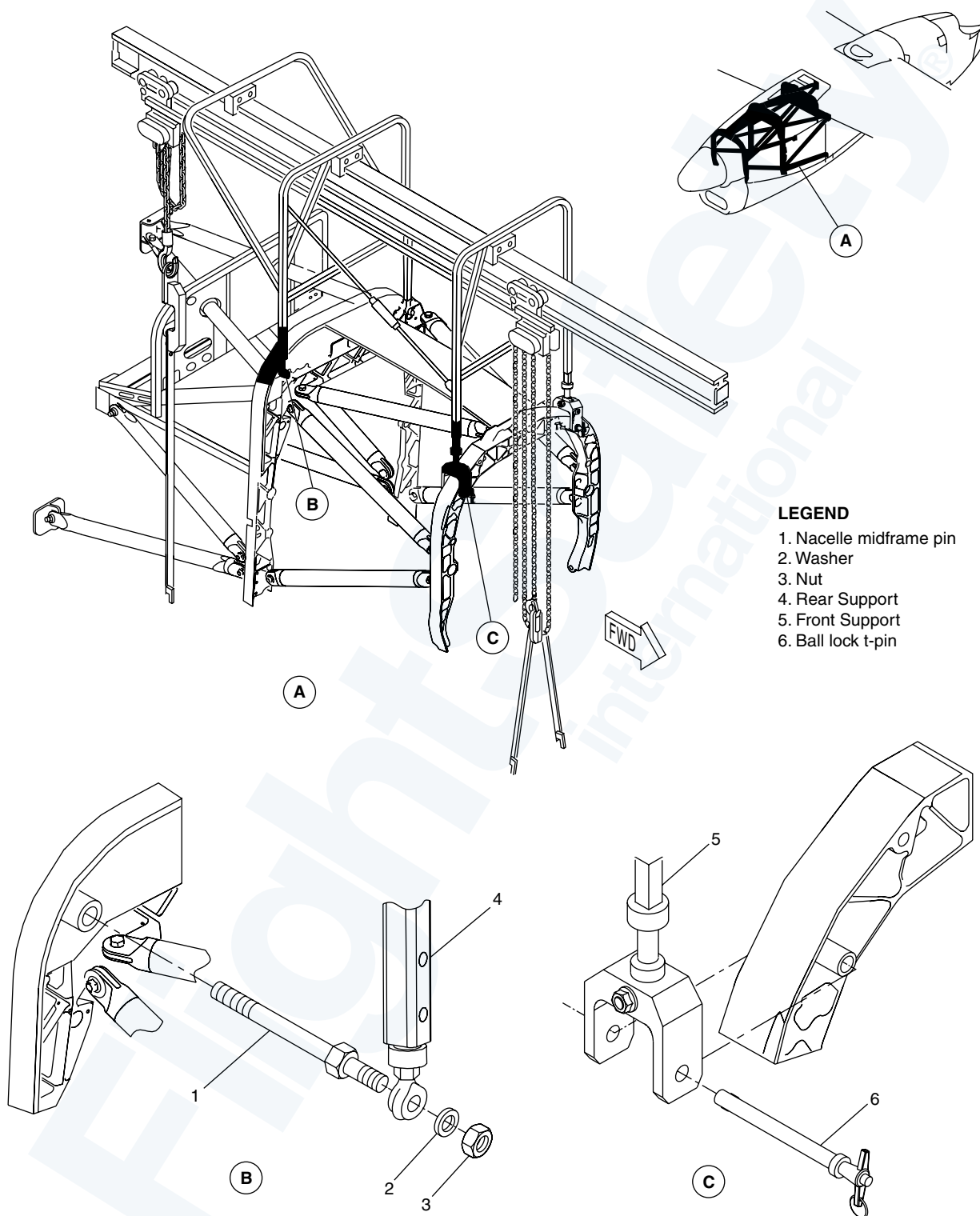
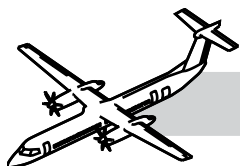
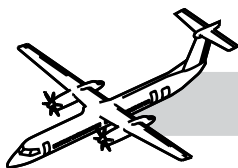


Figure 71-4. Engine Hoist - Components



INSTALLATION OF THE ENGINE HOIST

The following is an abbreviated description of the maintenance practice and is intended for training purposes only. For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.

The procedures that follow are for the installation of the Nacelle Mounted Engine Hoist.

WARNING

TORQUE THE NACELLE MOUNTED HOIST MOUNTING BOLTS TO THE SPECIFIED TORQUE.

Install the engine hoist GSB7100022 on the nacelle as follows:

1. Install the nacelle mid-frame pins to the nacelle frame.
2. Torque the aft 1/2 in. diameter mounting nut on the mid-frame pins to 85 lbf•ft)
3. Install the front support on the nacelle front frame and insert the Ball Lok-T pins
4. Install the rear support to the mid-frame pins with washers and nuts
5. Torque the 1/2 in. diameter nuts to 85 lbf•ft (115.25 N•m).
6. For the full installation of the hoist assembly, refer to the hoist manufacturer's operation and servicing manual.

Refer to:

- Figure 71-4. Engine Hoist - Components.
- Figure 71-5. Engine Change.

Install the engine hoist on the nacelle as follows:

- Remove the Fwd Cowling and Fwd Cowl Doors

- Install the engine hoist on the top of the nacelle and secure it with the nuts and bolts
- Torque the mounting bolts.

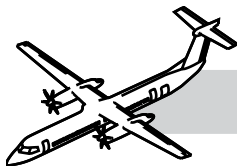
REMOVAL OF THE ENGINE HOIST

- Lower the engine and remove load from the engine sling
- Remove the engine sling from the engine
- Remove engine hoist.

For complete installation and removal of the hoist assembly, refer to the hoist manufacturer's operation and service manual.



Figure 71-5. Engine Change



INSTALLATION OF THE ENGINE IN THE NACELLE

The following is an abbreviated description of the maintenance practice and is intended for training purposes only. For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.

Make sure that the aircraft is in the same configuration as in the removal task.

CAUTION

MAKE SURE THAT THE INTAKE AIR PASSAGE IS CLEAR OF UNWANTED OBJECTS BEFORE INSTALLATION.

Prepare to install the engine in the nacelle as follows:

- Whether the engine is in the stand, or in a shipping container, position the engine under the nacelle approximately 12 inches forward of the installed position
- Attach the engine hoist to the forward and aft lifting points on the engine
- Install the rear lifting strut so that the cut-out faces aft

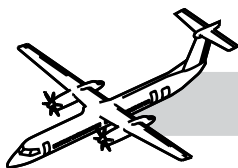
CAUTION

DO NOT LIFT THE ENGINE WHILE IT IS ATTACHED TO THE BASE OR THE STAND.

- With the hoist lift the engine from the stand or the shipping container
- Remove all the shipping closures from the engine
- Lift the engine to the correct height, then move it aft until the engine mount locations are aligned with the nacelle mount locations

- Install the isolators in the reverse order of removal
- Disconnect the hoist from the engine
- Connect the hydraulic pump
- Install the fuel lines on the FMU and the mid-frame
- Connect the oil cooler hoses
- Install the pneumatic ducts
- Install the electrical equipment and the loom connectors.
- Service the engine oil system with clean oil
- Install the propeller

Remove the engine hoist.



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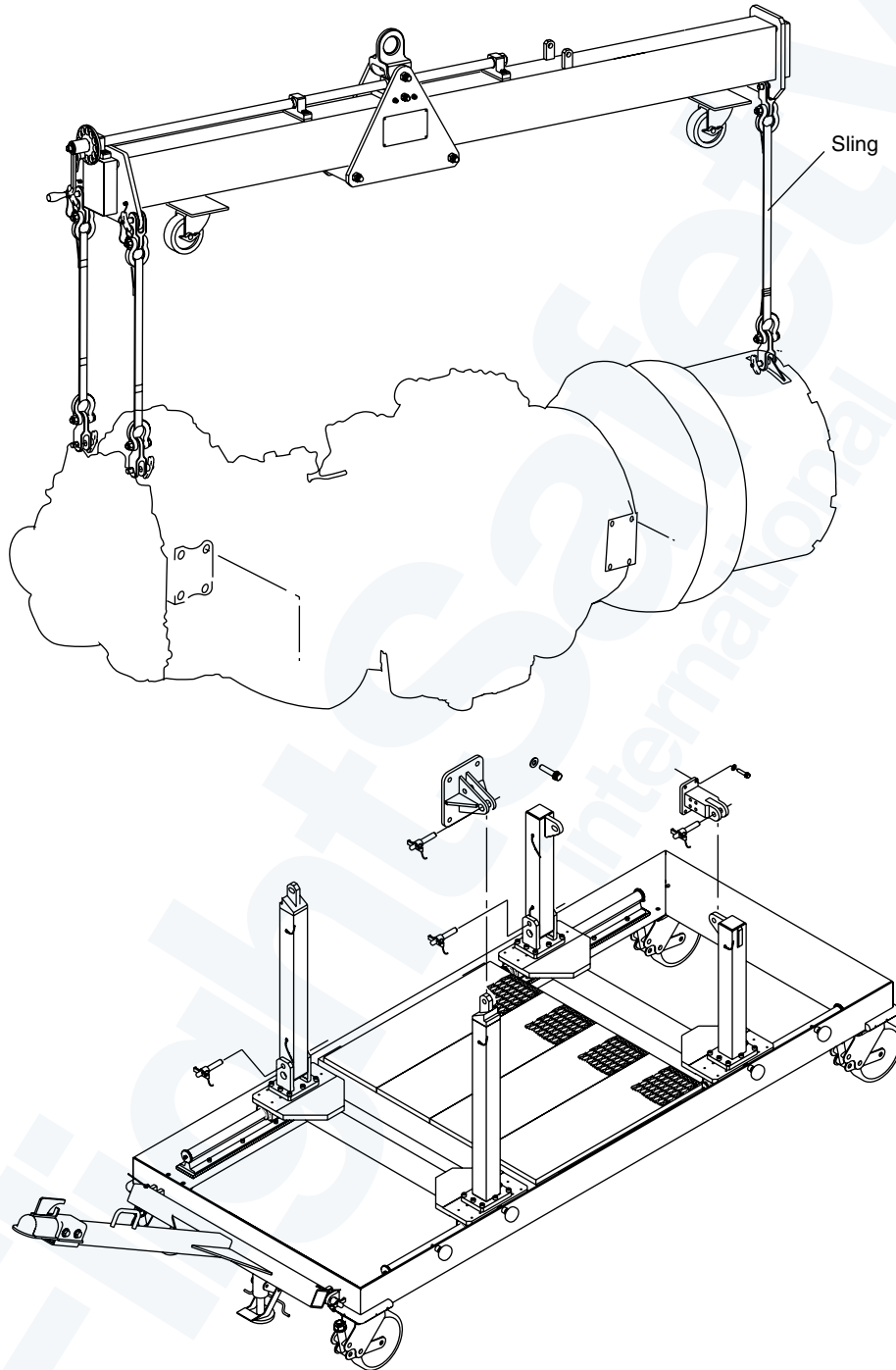
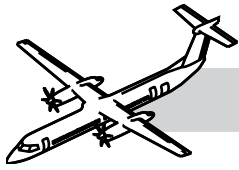
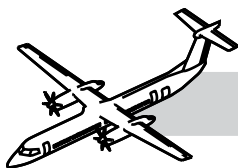


Figure 71-6. Engine Stand for Maintenance Practices



REMOVAL OF THE ENGINE FROM THE NACELLE

Refer to Figure 71-6. Engine Stand for Maintenance Practices.

This maintenance procedure is for the removal of the engine from the nacelle.

- De-energize the electrical system. Obey all the electrical/electronic safety precautions as well as all electrostatic discharge safety precautions
- Install the tail stand to the aircraft

NOTE

With both engines removed, tail stand installed and 1000 lbs (454.6 kgs) ballast added, the aircraft is safe to handle a maximum of 4 maintenance personnel in the aft baggage compartment, OR six maintenance personnel at the aft entry door.

No other maintenance personnel are allowed in the aircraft Aft of Fuselage Station X400.00.

NOTE

With a single engine removed, tail stand installed and 500 lbs (227.3 kgs) ballast added, the aircraft is safe to handle a maximum of 4 maintenance personnel in the aft baggage compartment, OR six maintenance personnel at the aft entry door.

No other maintenance personnel are allowed in the aircraft Aft of Fuselage Station X400.00.

Maximum ballast weight on each pilot's seat is 150 lbs (68 kgs) and maximum flight compartment floor loading is 300 lbs (136 kgs) uniformly distributed over the entire floor area. For maximum cabin floor loading, refer to the Cargo Loading Manual (PSM 1-84-8A).

- Open and tag the circuit breakers as per the AMM. Then, remove the lower cowl, access doors and cowls. Pull the T-handle (FUEL/ HYD shut-off).

Complete the following steps:

- Remove the propeller
- Install the engine hoist on the nacelle
- Attach the engine hoist to the forward and aft lifting points on the engine
- Install the rear lifting strut so that the cut-out faces aft
- Disconnect the electrical connectors from the engine
- Disconnect or remove the pneumatic ducts from the engine
- Remove or disconnect the electrical equipment
- Disconnect the oil, fuel and hydraulic systems.

Remove the isolators in the order that follows:

- The aft left isolator
- The aft right isolator
- The forward left isolator
- The forward right isolator
- The top isolator.

NOTE

It is easier to remove the rear isolators if you remove the center bolt first, then disassemble each isolator.

Move the engine forward approximately 12 in. Then, lower the engine and install it in the stand.

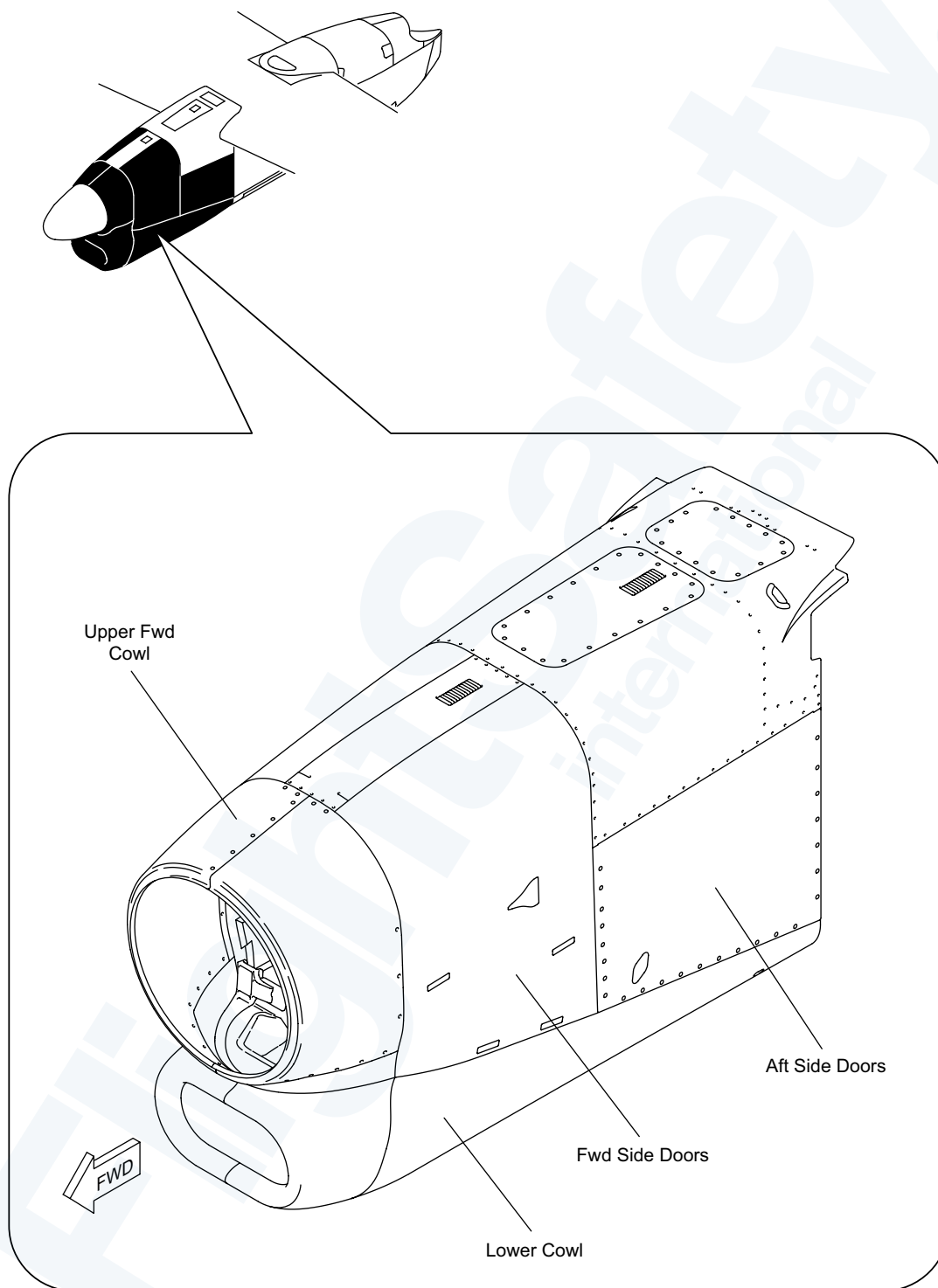
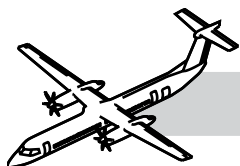
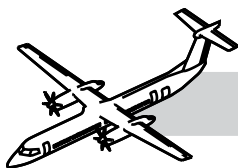


Figure 71-7. Cowling Locator and Details



71-10-00 COWLING

INTRODUCTION

The forward cowling forms part of the nacelle structure. It provides aerodynamic contour to the engine, and protects the Engine Mounting Structure (EMS).

GENERAL

The forward nacelle encloses:

- Engine
- Mounting structure
- Gearbox
- Engine systems.

SYSTEM DESCRIPTION

Refer to:

- Figure 71-7. Cowling Locator and Details.
- Figure 71-8. Nacelle Cowlings.

The cowling doors give access to the engine and components for maintenance.



Figure 71-8. Nacelle Cowlings

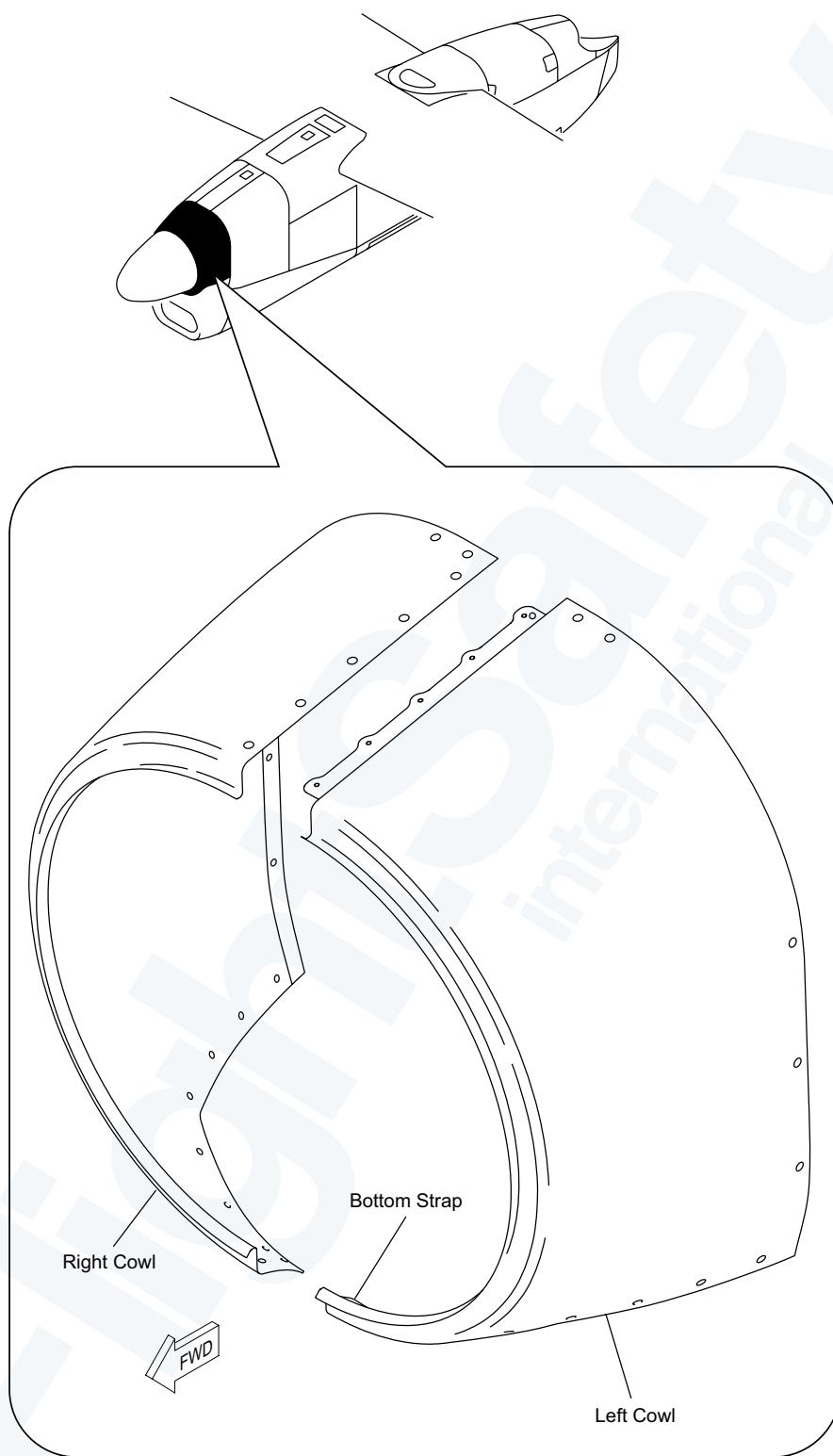
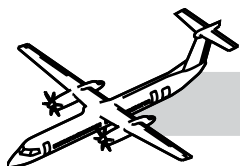
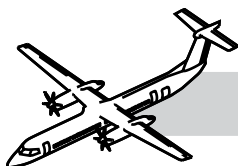


Figure 71-9. Upper Forward Cowl



COMPONENT DESCRIPTION

NOTE

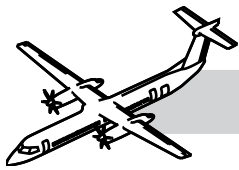
Upper Forward Cowl

Refer to Figure 71-9. Upper Forward Cowl.

The forward cowl is behind the spinner and covers the propeller shaft and the forward part of the reduction gear box.

The forward cowl consists of two “halves” that are joined longitudinally at the top by a titanium buttstrap. The buttstrap is permanently fastened to the left half and joined to the other by steel fasteners. The “halves” are manufactured from carbon-epoxy.

There is a gap between the front edge of the forward cowl and the spinner to allow cooling air to enter the firezone.



LEGEND

- 1. Duct DC Generator
- 2. Forward Side Door
- 3. Door Strut
- 4. Duct DC Generator
- 5. Door Latches
- 6. Spine Assy. (Ref)

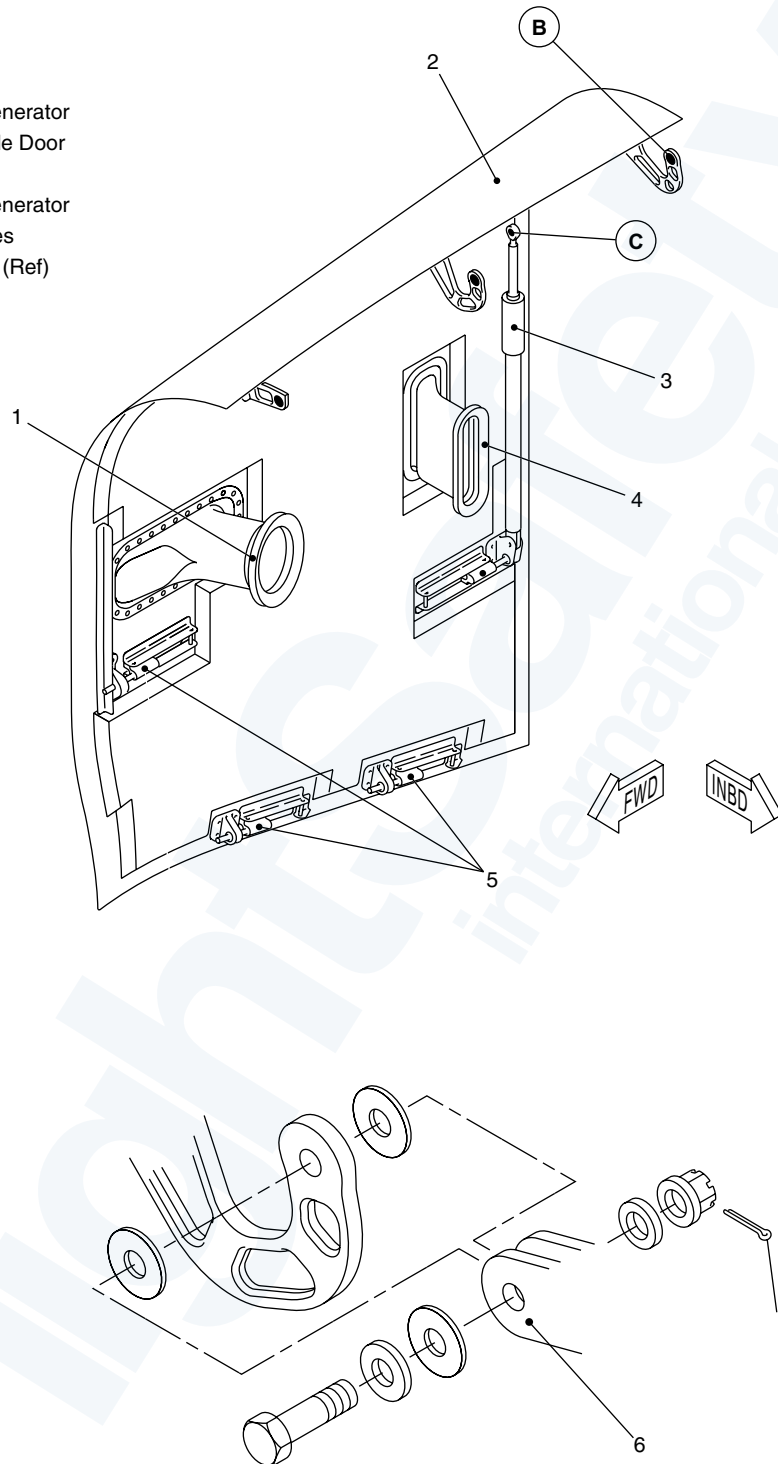
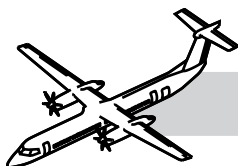


Figure 71-10. Forward Side Door



Forward Side Doors

Refer to:

- Figure 71-10. Forward Side Door.
- Figure 71-11. Engine Access Fwd Side Door Open.

There is a forward side door on each side of the nacelle immediately aft of the forward cowl.

The doors consist of carbon-epoxy skins with a central core of aluminum honeycomb.

The doors allow access to the engine and system forward of the mid frame. Each door is attached via three “goose neck” hinges to the 0.040 inch thick titanium spine that runs along the top of the nacelle.

Each door can be opened through a full 90 degrees and secured in the open position by a telescoping strut at the front corner of the door. The struts are self-locking in the half open and full open positions.

Each door is held closed by 4 latches; 2 on the lower edge and 1 each on the forward and aft edges. The door edges are sealed by a silicone rubber P-seal.

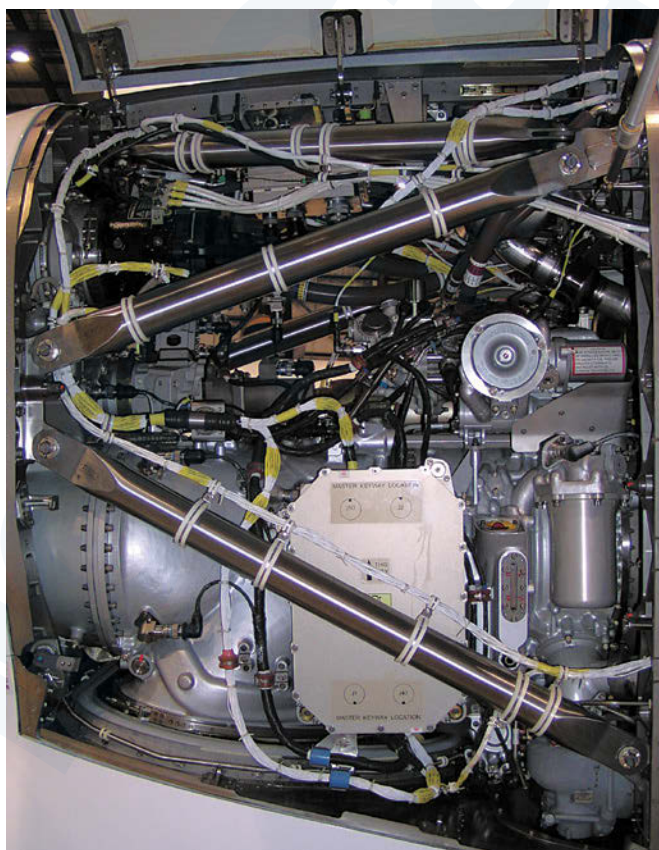
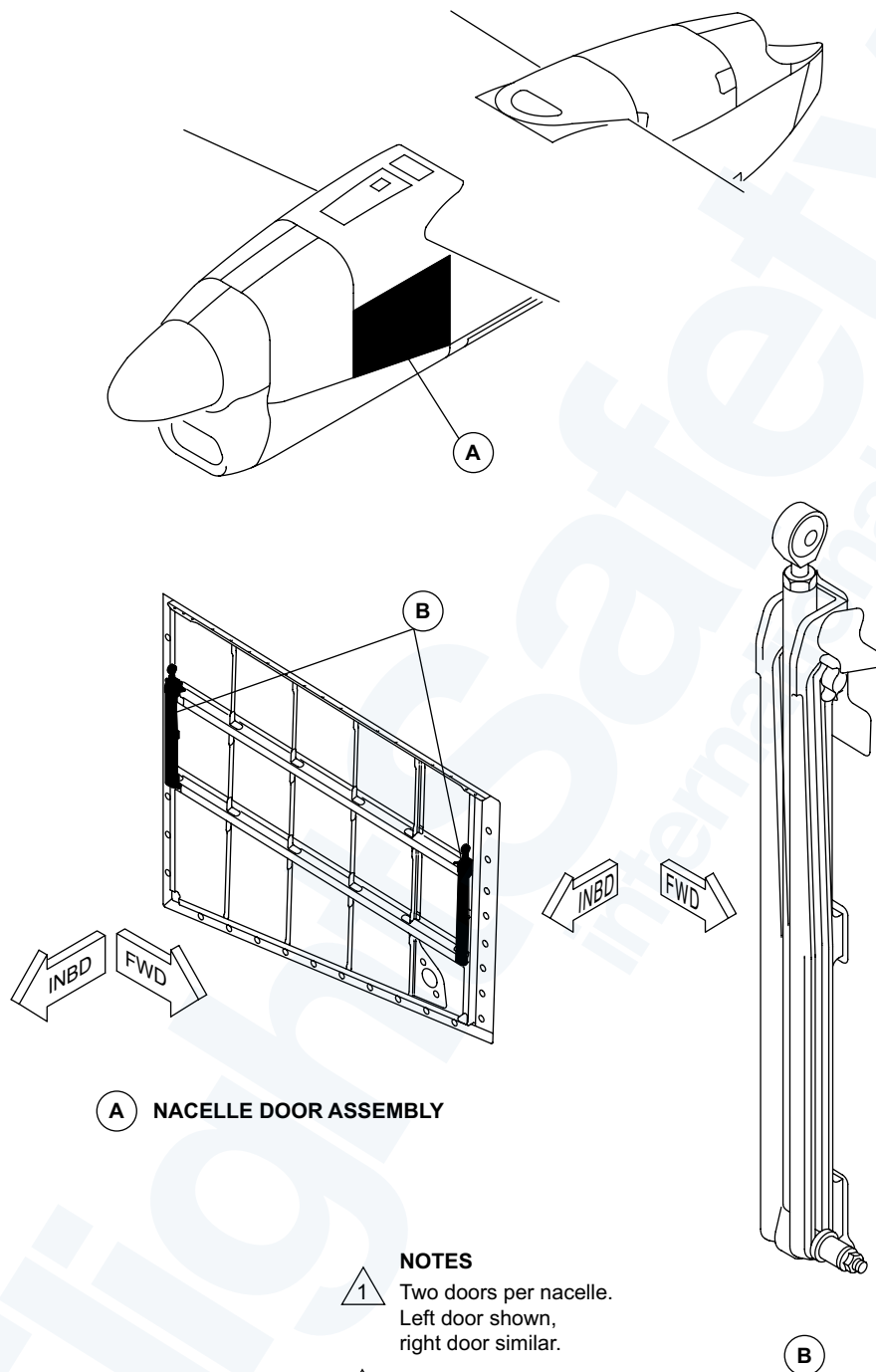
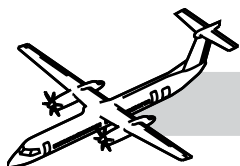


Figure 71-11. Engine Access Fwd Side Door Open

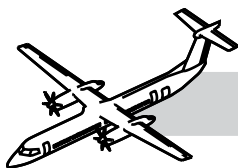


(A) NACELLE DOOR ASSEMBLY

NOTES

- 1 Two doors per nacelle.
Left door shown,
right door similar.
- 2 Aft strut shown,
Fwd strut similar.
- 3 Left nacelle shown,
right nacelle similar.

Figure 71-12. Aft Side Doors



Aft Side Doors

Refer to:

- Figure 71-12. Aft Side Doors.
- Figure 71-13. Engine Access Aft Side Door Open.

The aft side doors are below the leading edge fairing and aft of the forward side doors.

The aft side doors are manufactured from titanium. Each door is hinged at the top edge to the leading edge fairing by a “piano hinge” and locked closed by 25 quick release fasteners that are spaced evenly around the door edges.

The aft side doors enclose the area of the fire zone that is below the leading edge fairing and aft of the forward side doors. Each door has a strut at the forward and rear edges to hold the doors open during maintenance.

The struts are self locking in the “door open” position. When the doors are opened, access is provided to the rear engine mounts and struts, borescope ports, and rear firezone system.

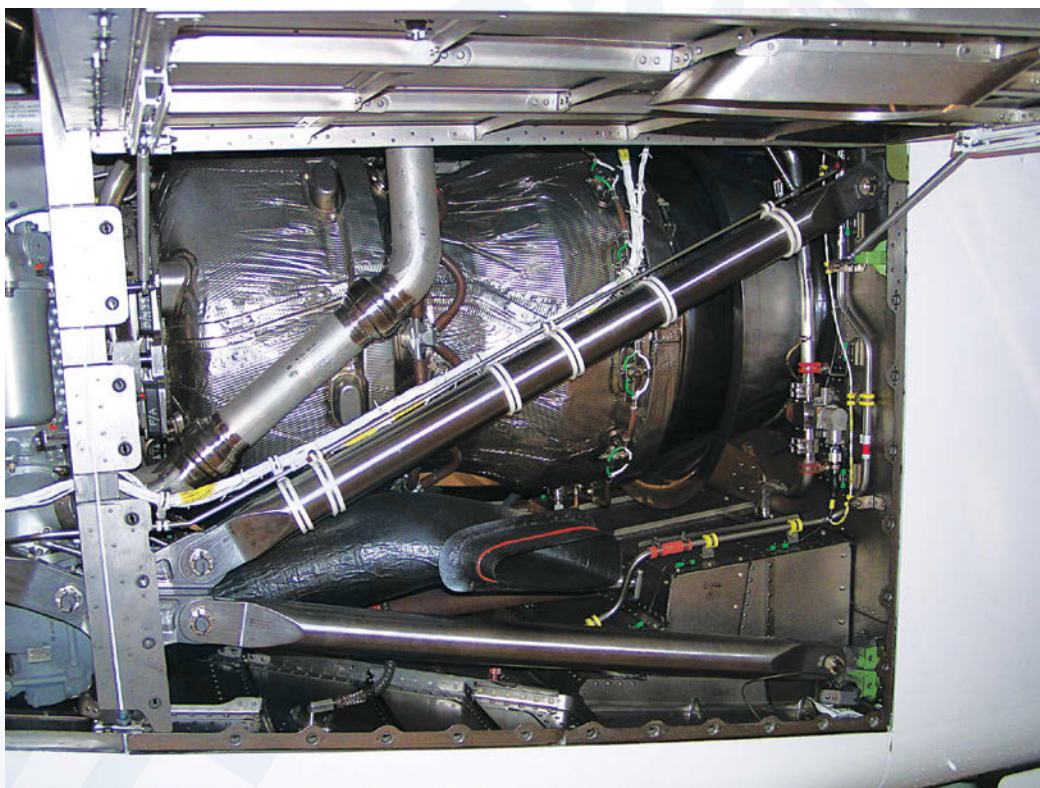


Figure 71-13. Engine Access Aft Side Door Open

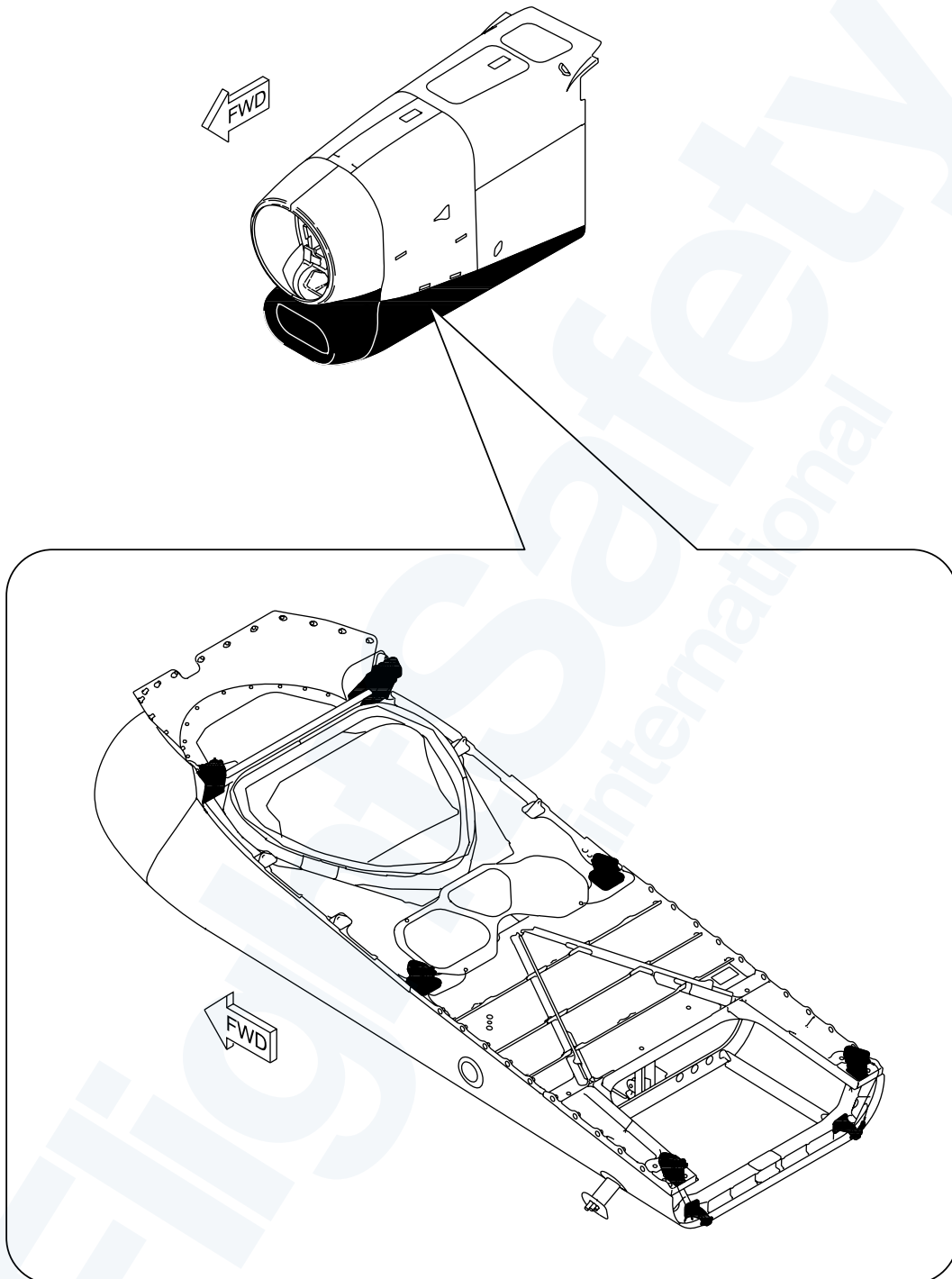
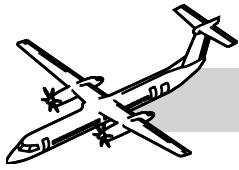
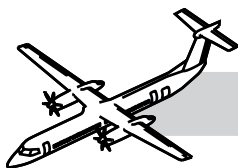


Figure 71-14. Lower Cowl



Lower Cowl

Refer to:

- Figure 71-14. Lower Cowl.
- Figure 71-15. Lower Cowl.

The lower cowl is at the bottom of the nacelle.

The outer skin of the lower cowl and the intake duct are moulded from carbon-epoxy/nomex honeycomb laminate. The upper surface of the lower cowl is a machined titanium structure with a minimum thickness of 0.050 inches and forms the “firefloor”.

The lower cowl contains:

- Air induction system
- Oil cooler system
- Inlet lip pneumatic de-icing system
- Firezone drain lines and drain mast
- Wing ice inspection light.

The lower cowl attaches to the nacelle by a hinge on either side of its aft end. The nacelle hinge point is just forward of the firewall structure.

The cowl is latched in place by 6 expandable tight fitting pins, and can be swung down to a maintenance position to allow access to the lower side of the engine. The lower cowl can be removed from the nacelle by pulling out the two hinge pins.



Figure 71-15. Lower Cowl

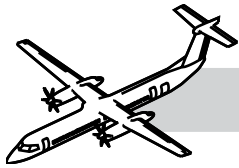


Figure 71-16. Expansion Bolt Open



Figure 71-17. Expansion Bolt Closed

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Figure 71-18. Fwd Expansion Bolt

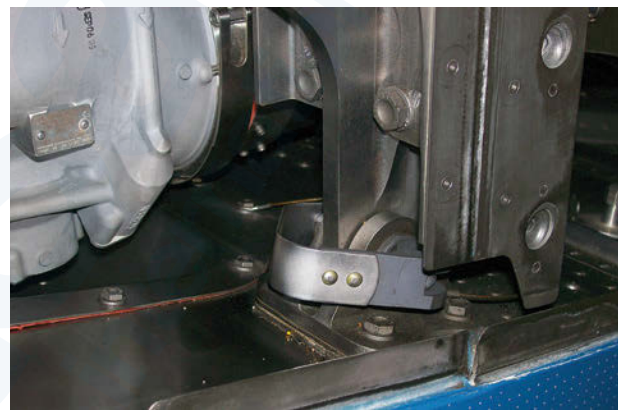


Figure 71-19. Mid Expansion Bolt

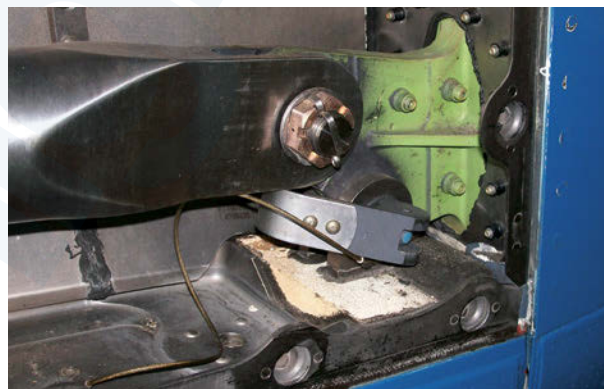
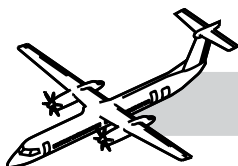


Figure 71-20. Aft Expansion Bolt



REMOVAL OF THE LOWER COWL

Remove the forward upper cowl

Refer to:

- Figure 71-16. Expansion Bolt Open.
- Figure 71-17. Expansion Bolt Closed.
- Figure 71-18. Fwd Expansion Bolt.
- Figure 71-19. Mid Expansion Bolt.
- Figure 71-20. Aft Expansion Bolt.
- Figure 71-21. Sling - Lower Fwd Cowl.

Open the forward side doors as follows:

NOTE:

Make sure that the door strut is correctly locked.

Open the aft side doors as follows:

NOTE:

Make sure that the door struts are correctly locked.

CAUTION

SET THE PROPELLER BLADES AT APPROXIMATELY 45 DEGREES FROM THE VERTICAL CENTER LINE BEFORE YOU LIFT (OR LOWER) THE FORWARD NACELLE COWL. IF YOU DO NOT DO THIS, YOU WILL CAUSE DAMAGE TO THE TRAILING EDGES OF THE PROPELLER BLADES.

CAUTION

THE LOWER COWL IS A PRIMARY STRUCTURE.

YOU MUST CORRECTLY SUPPORT THE LOWER COWL IF YOU REMOVE IT. IF YOU DO NOT DO THIS, YOU CAN CAUSE DAMAGE TO THE AIRCRAFT AND THE COMPONENT.

Remove the lower cowl as follows:

- Disconnect the engine drains
- Install the sling (ACS28004) on the lower cowl
- Remove the six expandable pins.

CAUTION

DO NOT OPEN THE LOWER COWL TO MORE THAN 45 DEGREES.

- Slowly, operate the sling and lower the front end of the lower cowl
- Move the transport stand (ACS28005) into position under the lower cowl
- Disconnect the electrical connectors
- Install the hoist (ACS28009)
- Remove the pip pins from the hinge assemblies
- Carefully lower the cowl onto the transport stand (ACS28005)
- Remove the hoist and the sling from the lower cowl
- Put an identification label on the lower cowl to identify its position
- Install the cover (GSB5411003) on the lower cowl.

CAUTION

DO NOT TOW THE LOWER COWL TRANSPORTATION TROLLEY AT SPEEDS MORE THAN 5 MPH (8 KM/H).

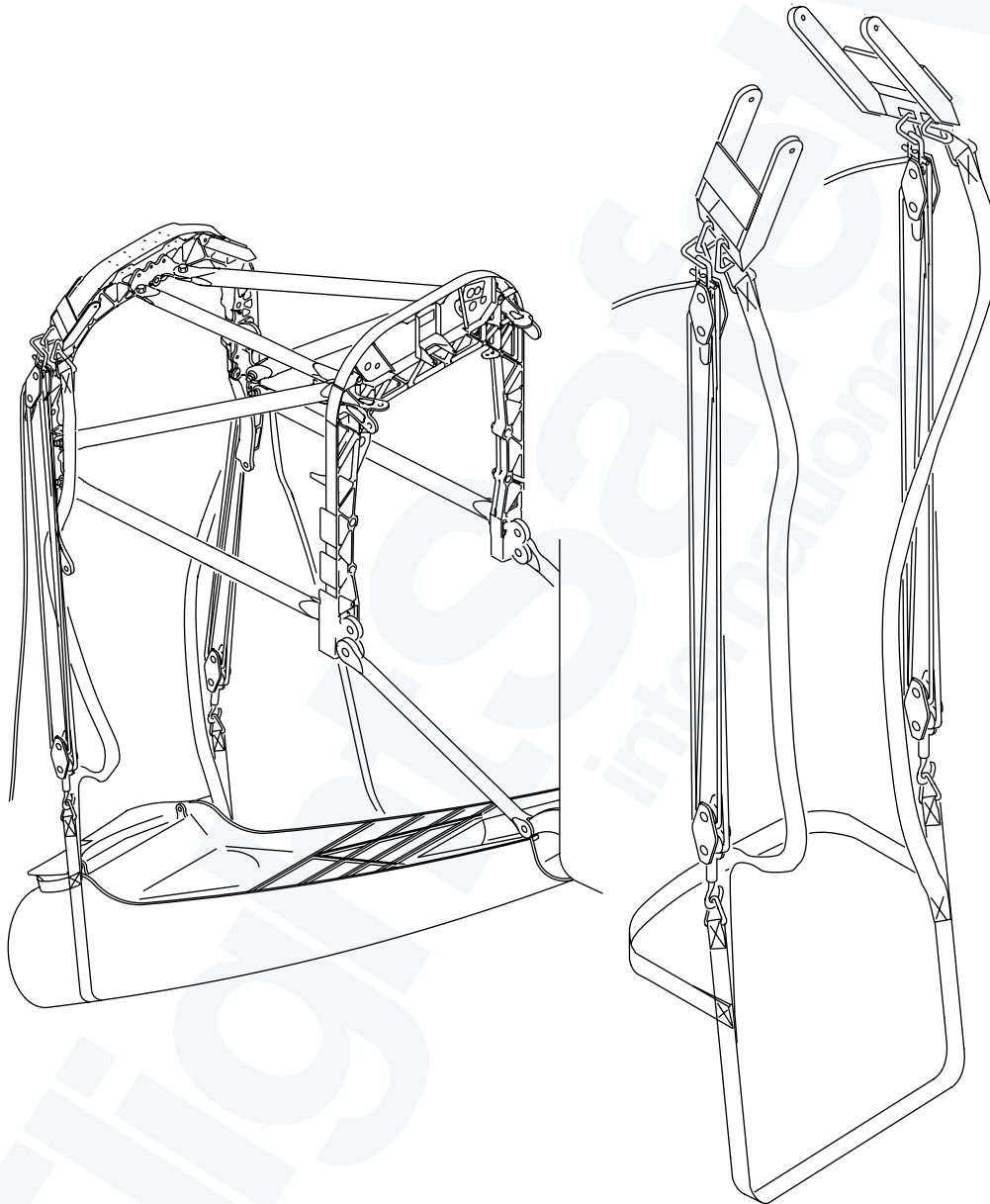
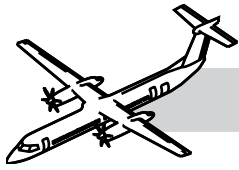
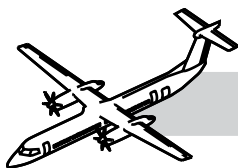
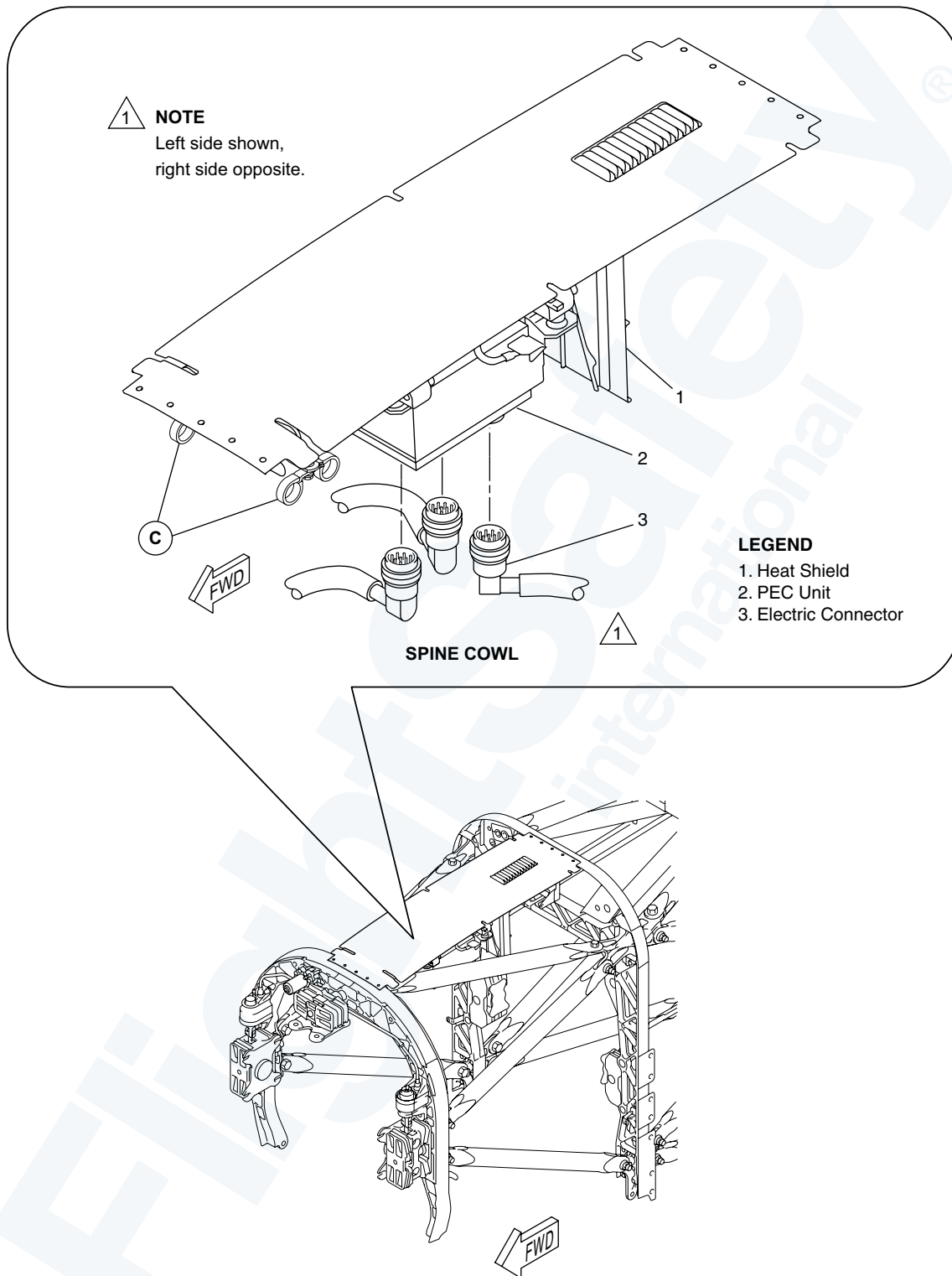
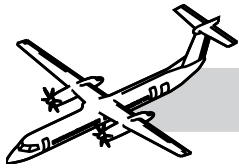
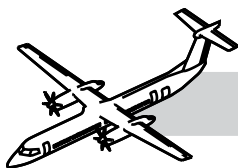


Figure 71-21. Sling - Lower Fwd Cowl



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Spine Cowl

NOTE

Refer to Figure 71-22. Spine Cowl.

The spine cowl assembly forms part of the engine mounting structure (EMS) and is bolted to the top of the nacelle between the forward and mid frames.

The spine cowl is a skin panel made of titanium.

The spine has three hinges on each side. These hinges support the forward side doors.

The spine also contains installation mounts for the propeller electronic controller.

A heat shield assembly for the pneumatic pre-cooler is bolted to the spine.

A louver in the skin (aft of the heat shield assembly) is the exhaust outlet for the pneumatic system pre-cooler.

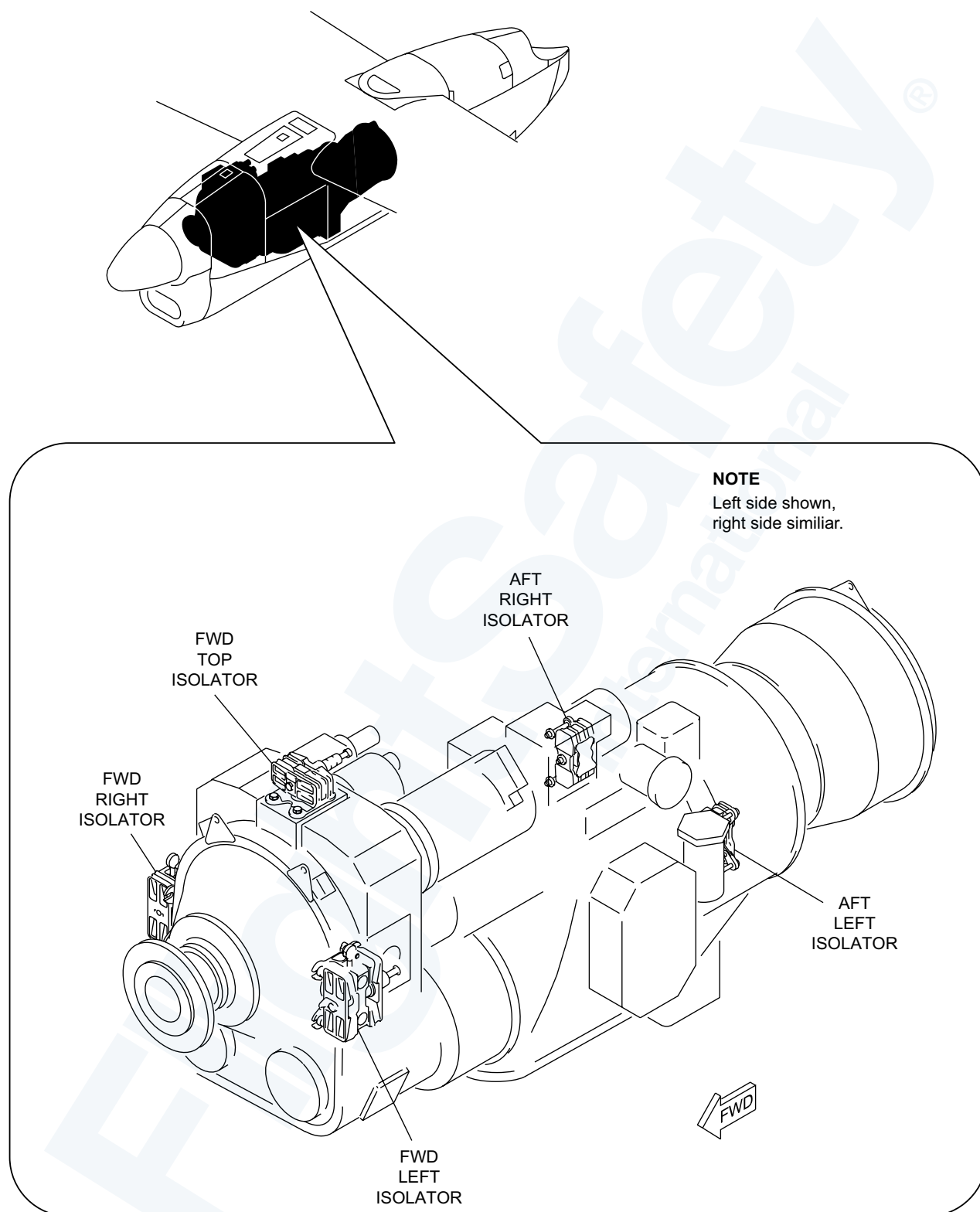
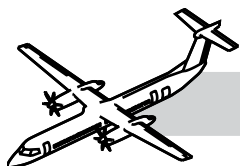
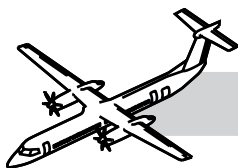


Figure 71-23. Engine Isolation System



71-21-00 ENGINE ISOLATION SYSTEM

NOTE

INTRODUCTION

The engine mounts support the engine and isolate torque and vibrations.

GENERAL

The engine mounting structure has a forward frame (horse collar) which supports the engine through three elastomeric vibration isolator mounts that transfers the loads from the propeller and engine into the struts through strut fittings.

The frame is a machined titanium section with an integral triangular web stiffening section, that is installed perpendicular to the engine axis, and is external of the engine rotor burst zone.

The machined titanium mid frame is used to attach the rear engine mounts.

SYSTEM DESCRIPTION

Refer to Figure 71-23. Engine Isolation System.

The PW150A engine has three front mount pads on the Reduction Gearbox (RGB), (two side and one top).

There are also two aft mount pads on the left and right sides of the Intercompressor Case.

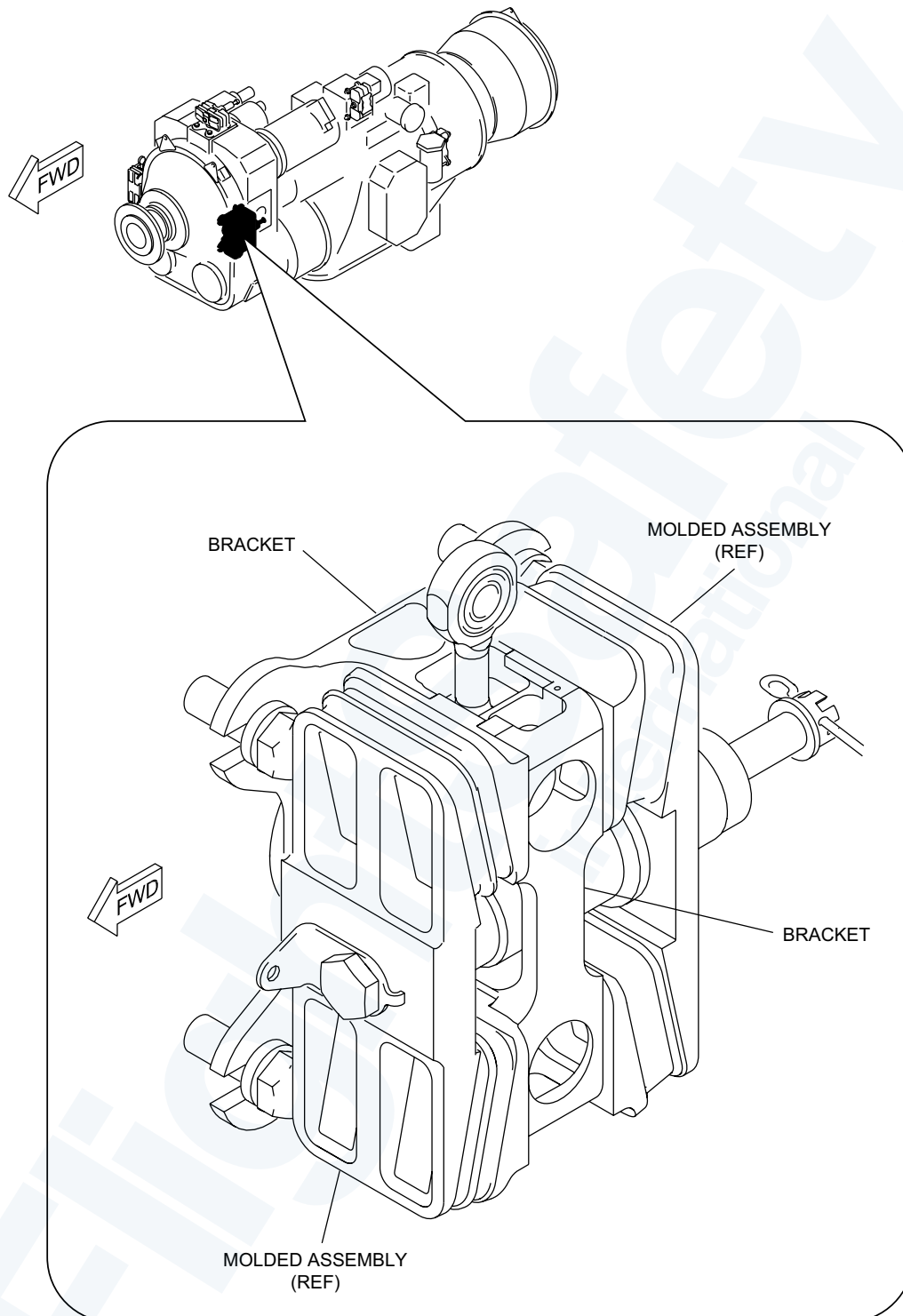
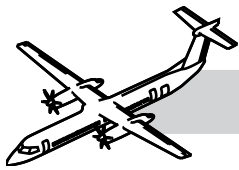
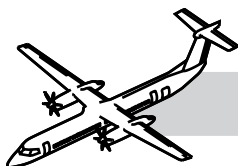


Figure 71-24. Forward Left and Right Isolator



COMPONENT DESCRIPTION

Forward Left and Right Isolator

Refer to:

- Figure 71-24. Forward Left and Right Isolator.
- Figure 71-25. Vibration Isolator Mount.

The forward side vibration isolators mount on the left and right forward side mount pads of the engine. They use identical hardware, and can be assembled in either a left or right configuration.

They react to axial, lateral, and vertical loads, and are linked to the HTCS for torque loads.

The mounts are redundant; i.e., system integrity is maintained after loss of one mount.

Clearance between the core and the bracket provides the snubbing envelope.

Each isolator has a bracket assembly and two isolator plates.

The isolator plates have a single plate with two elastomer pads bonded to one side.

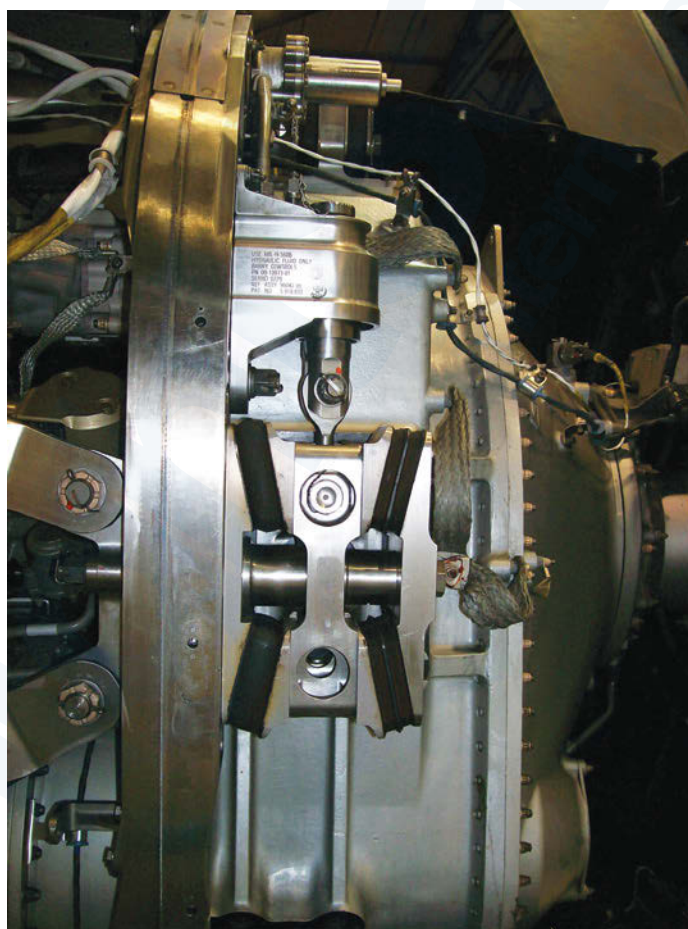


Figure 71-25. Vibration Isolator Mount

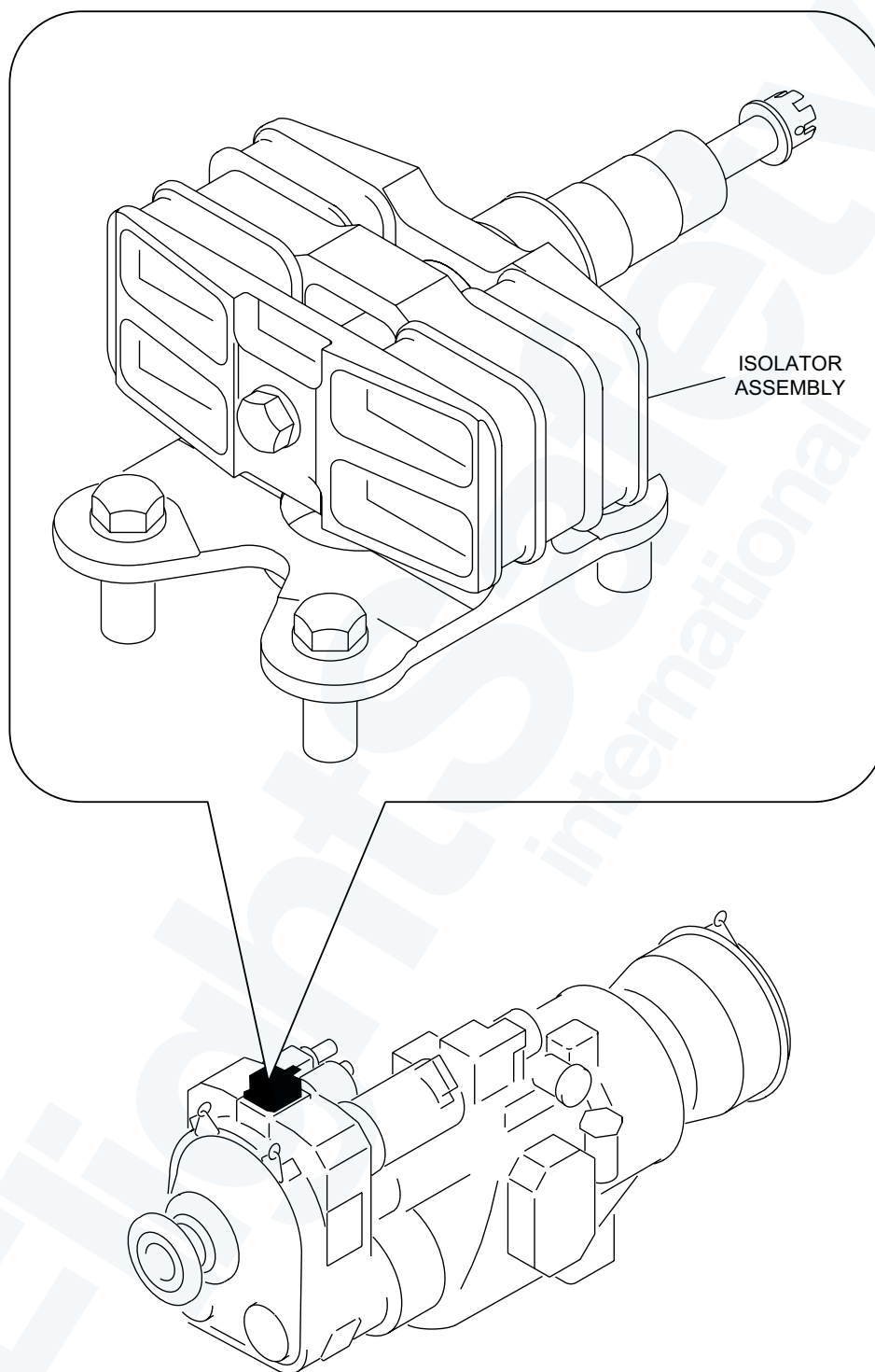
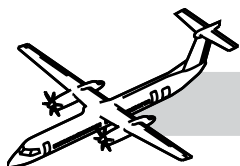
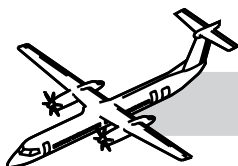


Figure 71-26. Forward Top Isolator



Forward Top Isolator

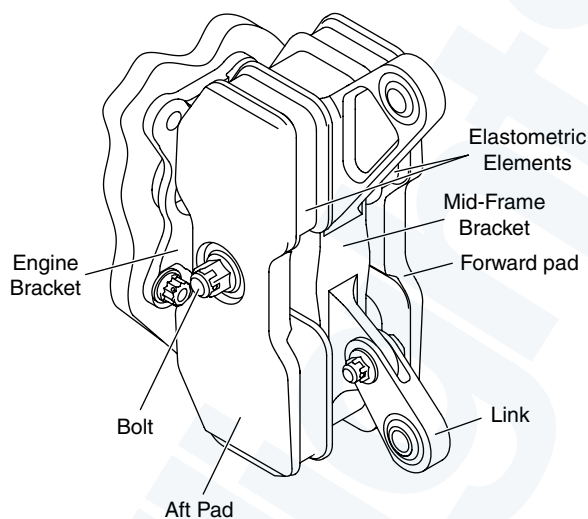
Refer to Figure 71-26. Forward Top Isolator.

The forward top vibration isolator mounts on the forward top mount pad of the engine.

It consists of a titanium bracket and two isolator plates with elastomer pads bonded to each inner surface. The forward plate is titanium and the aft plate is aluminum. The bracket is bolted to the engine mount pad by four stainless steel bolts. A single steel bolt and core assembly attach the plates to the fore and aft sides of the bracket, and fastens the complete assembly to the front frame. The bolt is secured by a nut and cotter pin.

The isolator provides engine restraint in the event that both aft mounts fail. Clearance between the core and the bracket provides the snubbing envelope.

The top vibration isolator should be installed first.



NOTE

Right mount shown, left mount similar.

Figure 71-27. Aft Left and Right Isolator

Aft Left and Right Isolator

Refer to:

- Figure 71-27. Aft Left and Right Isolator.
- Figure 71-28. Aft Side Mount.

The aft vibration isolators mount on the left and right aft mount pads of the engine.

They react to vertical and lateral engine loads, and a small percentage of the engine torque loads.

The isolator consists of an engine bracket, mid-frame bracket, two outer plates and a link. The outer plates have elastomer pads bonded to their inner surfaces.

The engine bracket is secured to the engine mount pad by four bolts. A single through bolt attaches spigots on the outer plates to a spherical bearing in the engine bracket.

The inner surface of the elastomer pads key to the faces of the mid-frame bracket. The mid-frame bracket attaches to the mid-frame at the top using a spherical bearing and a bolt, and at the bottom using the link, a spherical bearing and a bolt.

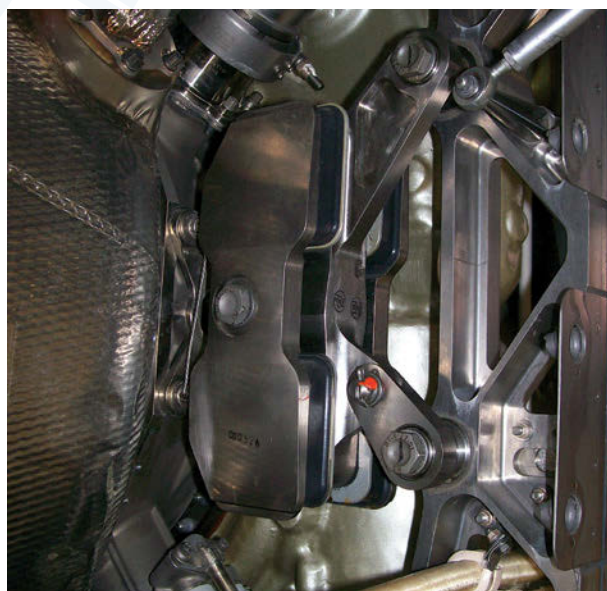


Figure 71-28. Aft Side Mount

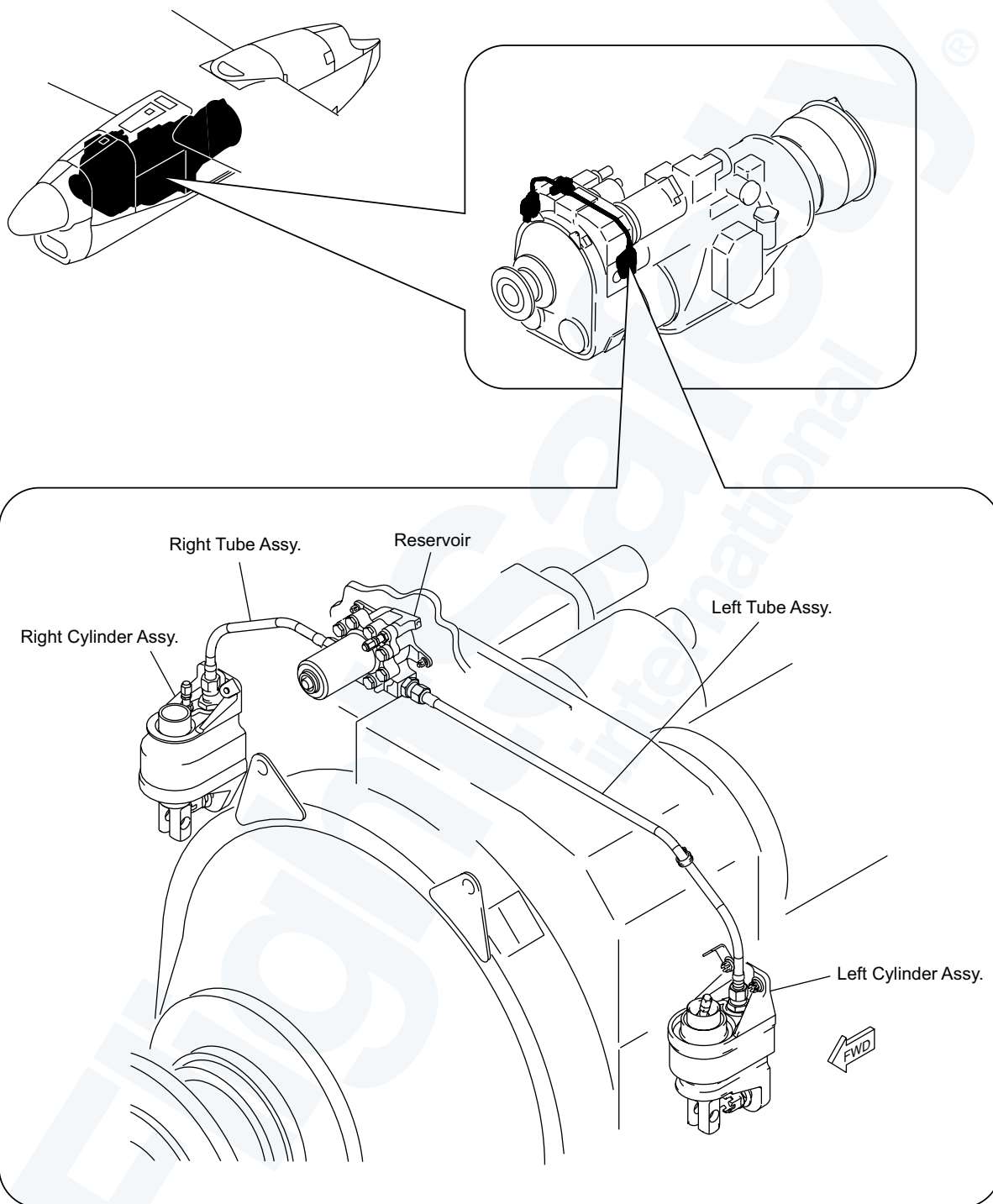
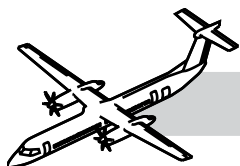
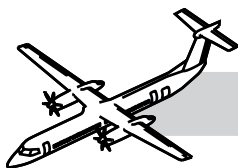


Figure 71-29. Hydraulic Torque Restraint System



71-22-00 HYDRAULIC TORQUE RESTRAINT SYSTEM

NOTE

INTRODUCTION

The Hydraulic Torque Compensation System (HTCS) provides high torsional stiffness that react to engine torque loads. It also provides low translational stiffness to minimize the transmission of vibrations.

GENERAL

The Engine Vibration Isolator System (EVIS) and HTCS work together, to provide a load path from the engine mount pads to the nacelle Engine Mount Structure (EMS).

SYSTEM DESCRIPTION

Refer to Figure 71-29. Hydraulic Torque Restraint System.

The HTCS is installed on the forward face of the front frame. The HTCS has:

- Two hydraulic actuator cylinders
- A hydraulic fluid reservoir
- Restrictor
- Check valve
- Connecting tubes.

The cylinder, reservoir, and supply tubes are made of stainless steel. The HTCS is qualified as fire resistant. In the unlikely event that a fire were to cause leakage of the hydraulic fluid, the front mounts will snub and react to the engine torque. In the event of a failure of the torque reaction system, the side mounts will snub and react to the torque.

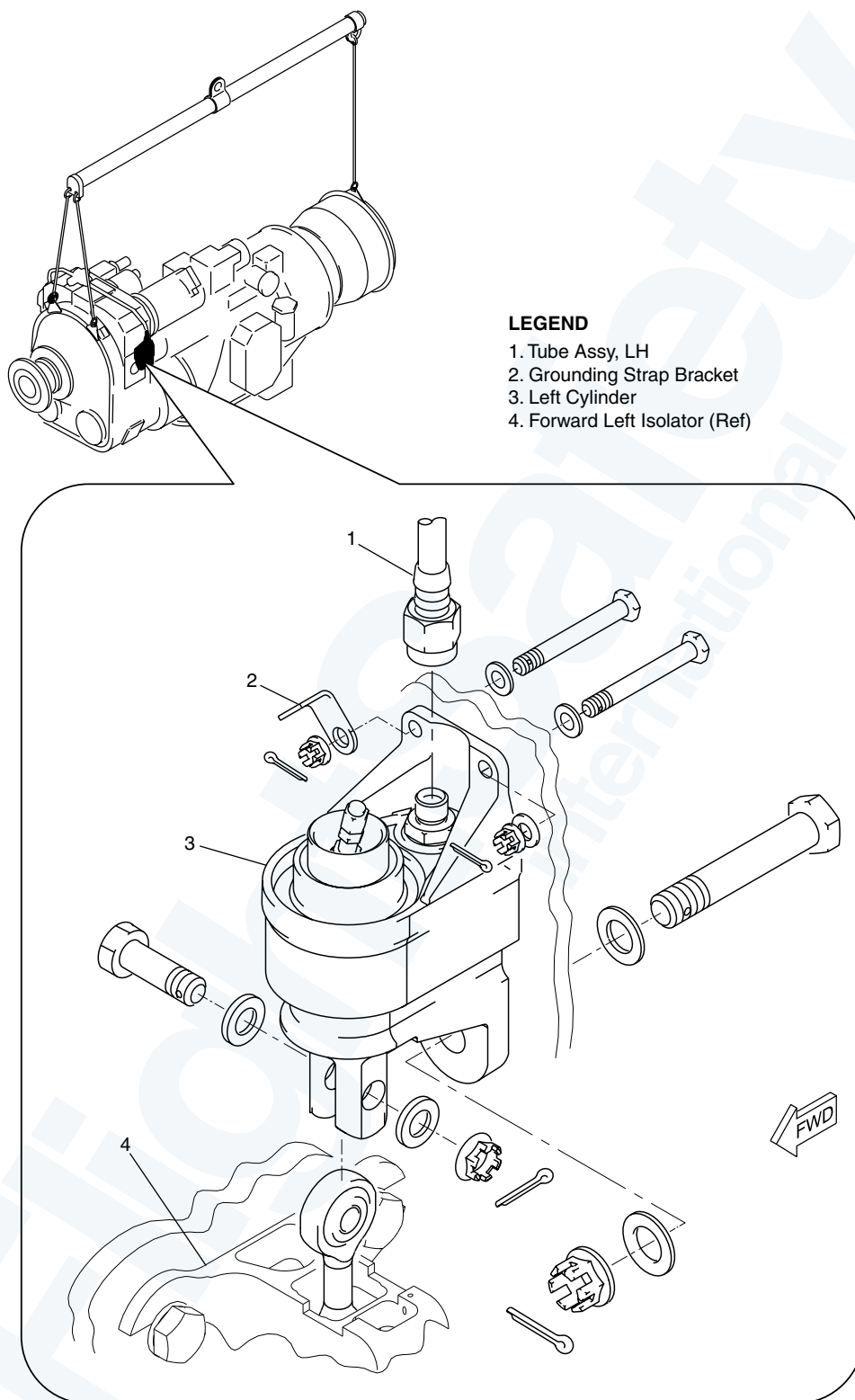
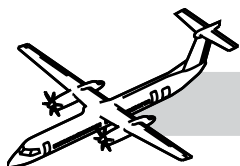
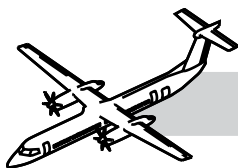


Figure 71-30. Cylinder Assembly



COMPONENT DESCRIPTION

Left and Right Cylinder Assembly

Refer to Figure 71-30. Cylinder Assembly.

The cylinder assemblies are installed on the left and right side of the engine support frame of No.1 and No.2 nacelle.

The cylinders are hydraulic actuators made of stainless steel.

Engine torque reacting on the rod end of the actuators compresses the hydraulic fluid. The hydraulic fluid is forced through a restrictor then compressing the spring in the reservoir.

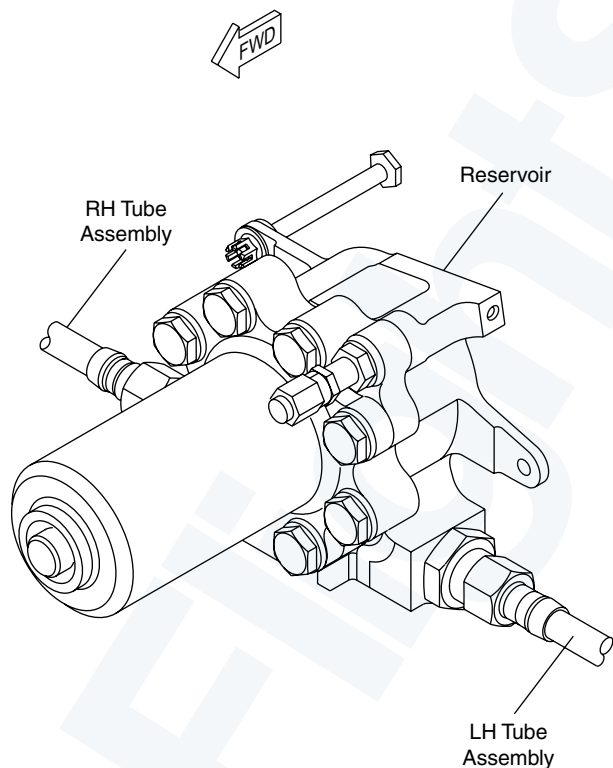


Figure 71-31. Reservoir and Tube Assembly

Hydraulic Torque Compensation - System Reservoir

Refer to:

- Figure 71-31. Reservoir and Tube Assembly.
- Figure 71-32. HTCS Reservoir.

The reservoir is mounted to the forward face of the front frame of No.1 and No.2 nacelle.

The hydraulic reservoir is aluminum alloy, and connecting tubes are made of stainless steel.

Hydraulic pressure is generated by the engine torque at the actuators. The hydraulic fluid is forced through a restrictor that compresses the spring in the reservoir.

FUNCTIONAL TEST OF HTCS RESERVOIR LEVEL INDICATOR

The following is an abbreviated description of the maintenance practice and is intended for training purposes only. For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.

- Measure and record the indicator extension
- Make sure extension is within limits stated in Task sheet.

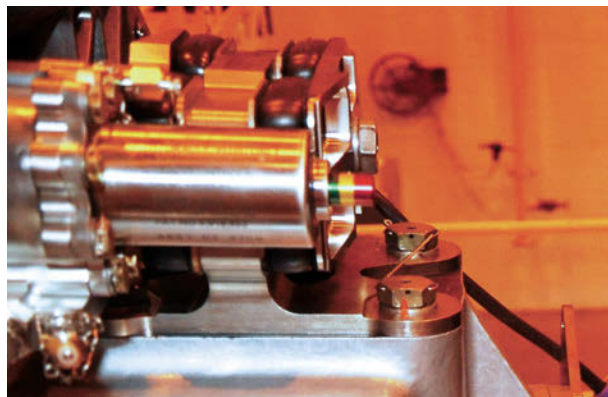


Figure 71-32. HTCS Reservoir

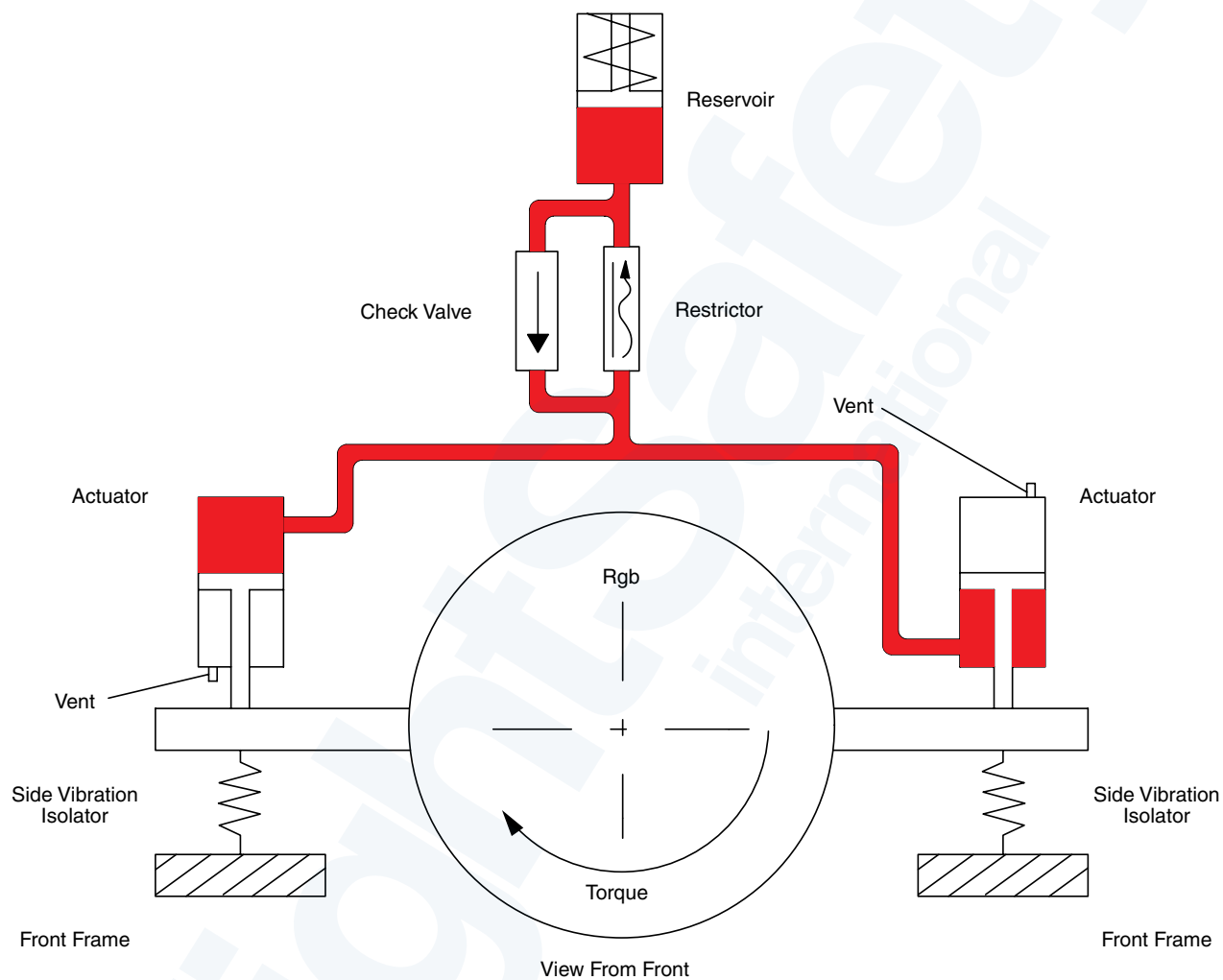
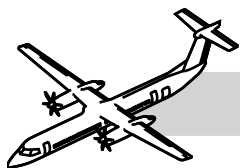
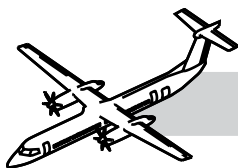


Figure 71-33. HTCS Operation



OPERATION

NOTES

Refer to Figure 71-33. HTCS Operation.

Normal System Operation

The hydraulic torque restraint system operates under a counter-clockwise positive torque load, when viewed from the rear. The actuators compress the hydraulic fluid into the reservoir through an orifice. A high torsional stiffness is created by the trapped fluid.

The system has no stiffness in a clockwise, (negative torque) load. Negative torque will be absorbed mainly by the side and partly by the rear isolators into the engine mounting structure. In this condition, fluid will transfer rapidly from the reservoir into the actuators through the check valve. A high negative torque may rotate the engine to the snubbing limit of the forward side isolators. At the snubbing limit, the HTCS actuators will not bottom out.

When the system has a vertical load, the HTCS acts as a hydraulic damper by transferring fluid from one actuator to the other. At low frequencies and small displacements, there is insignificant vertical damping.

The system contains approximately 12 cu.in. (165ml) of MIL-H 5606 hydraulic fluid. It is pre-pressurized against the pressure of the spring-loaded piston in the reservoir.

This is to avoid the cavitation during a rapid movement to negative torque. The pressure (i.e. volume of fluid) may be determined by measuring the stand-out of an indicator at the end of the reservoir.

The nominal system pressure at take-off power is 3,140 psig, including pre-pressure.

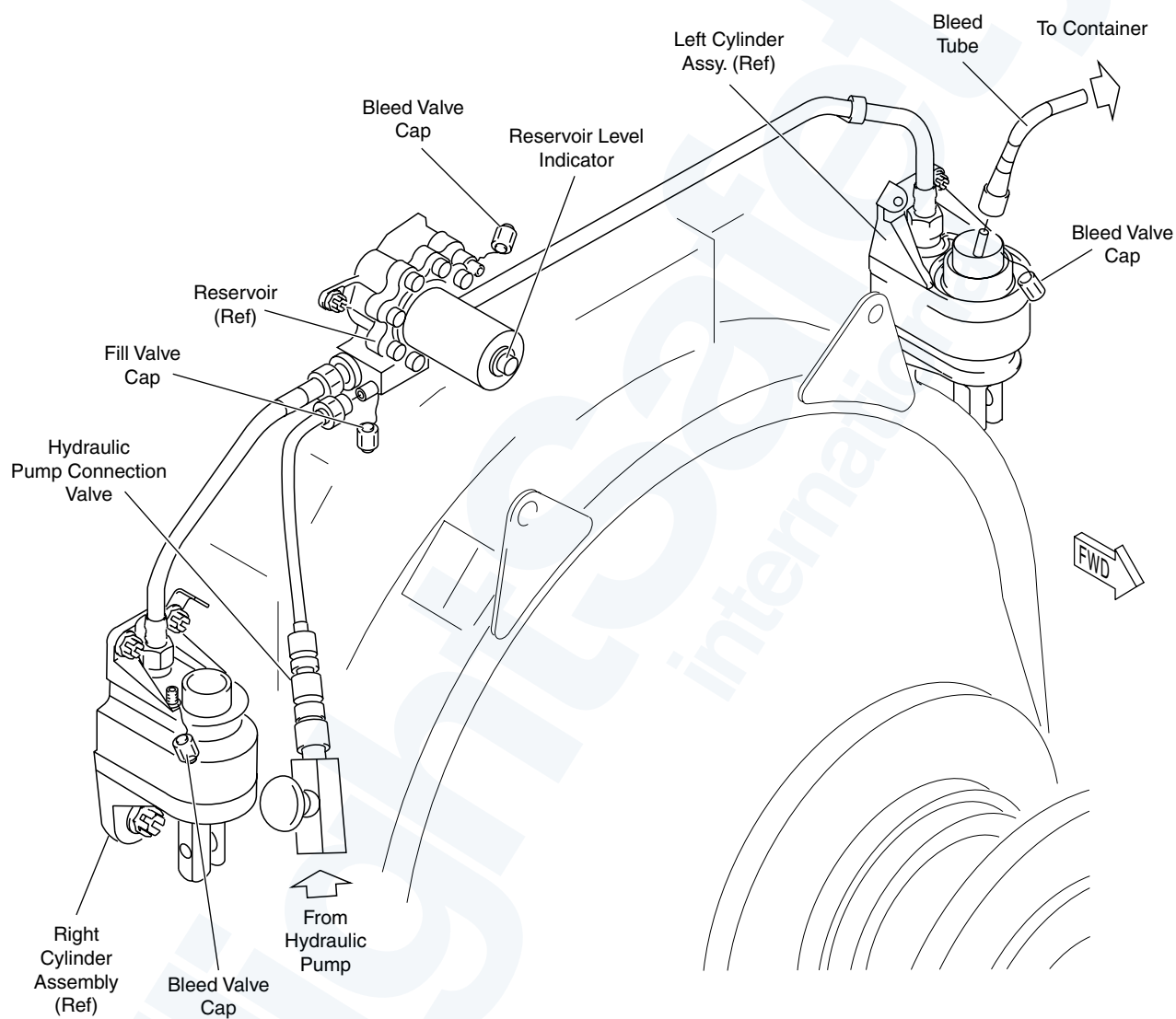
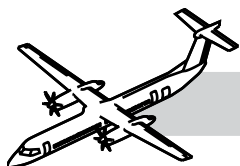
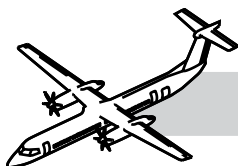


Figure 71-34. HTCS Servicing



SERVICING OF THE HTCS

NOTES

Refer to the Bombardier AMM PSM 1-84-2 for a detailed description of this maintenance practice.

The following is an abbreviated description of the maintenance practice and is intended for training purposes only. For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.

Refer to Figure 71-34. HTCS Servicing.

- Release the pressure from the system by removing the bleed caps from the cylinder assemblies and connecting the bleed tubes.
- Remove the bleed cap from the reservoir. Connect a bleed tube to the reservoir and to the hydraulic pump.
- Pump fluid into the reservoir until an air free flow is observed.
- Repeat procedure for the left and right cylinder assemblies.
- Release handpump pressure from the system and check level indication extension is correct.
- Install bleed caps
- Observe hydraulic fluid precautions during task.

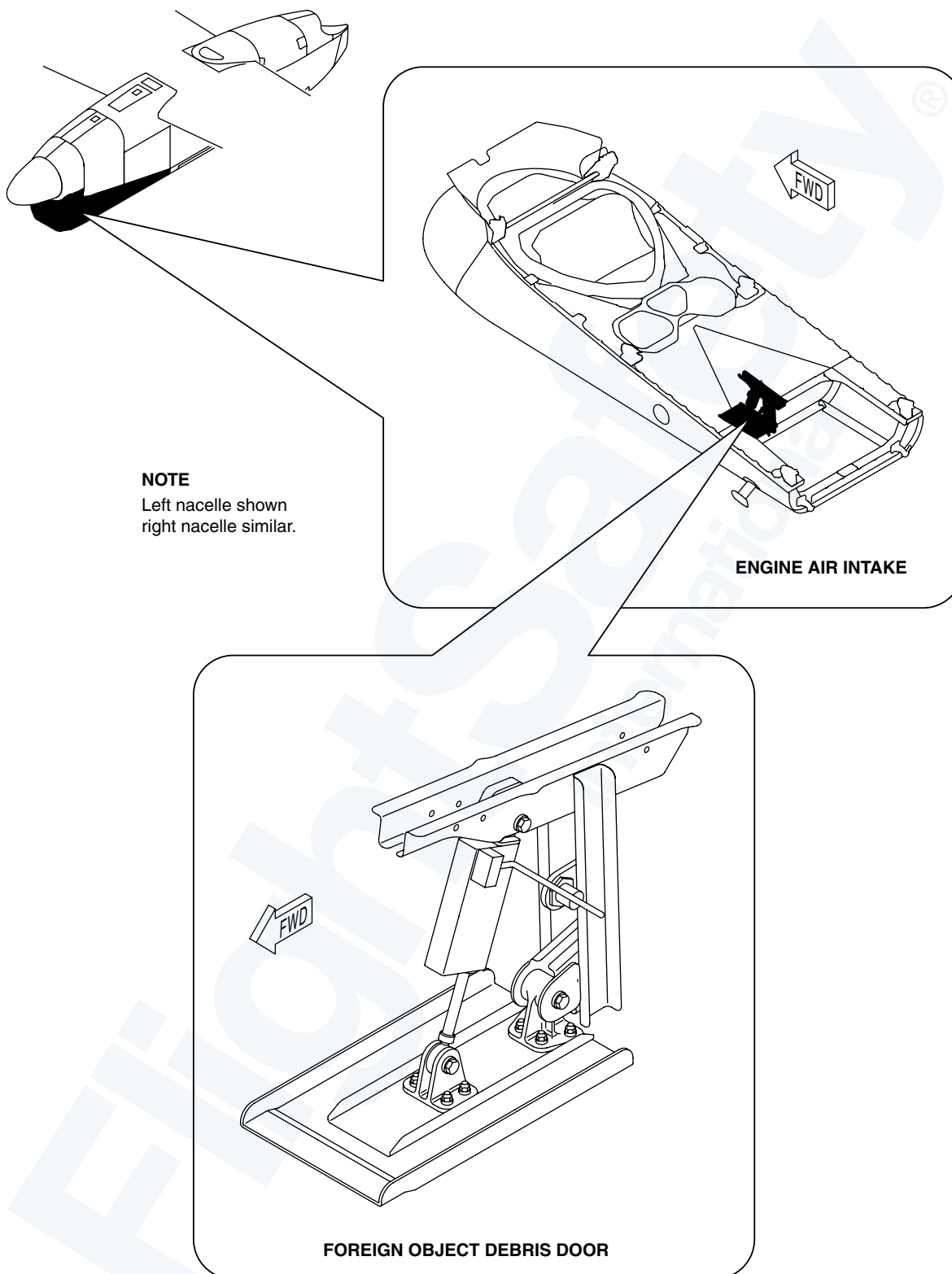
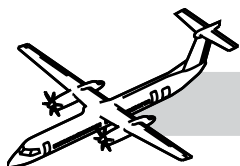
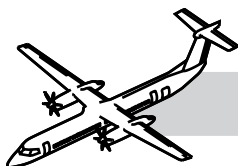


Figure 71-35. Forward Nacelle Air Intake



71-61-00 FORWARD NACELLE AIR INTAKE

INTRODUCTION

The air intake is part of the lower nacelle cowl.

GENERAL

Refer to Figure 71-35. Forward Nacelle Air Intake.

The air intake directs the air entering the engine for all engine operations. A Foreign Object Debris (FOD) door is installed at the rear of the duct, when the door is open debris is ejected out through the bottom surface of the lower cowl.

SYSTEM DESCRIPTION

Refer to:

- Figure 71-35. Forward Nacelle Air Intake.
- Figure 71-36. Forward Nacelle Air Intake Detail.

When the aircraft is in icing conditions and the deicing system is selected on, ice that is shed from the deicing boot could be ingested into the engine causing damage. This system will prevent this from happening.

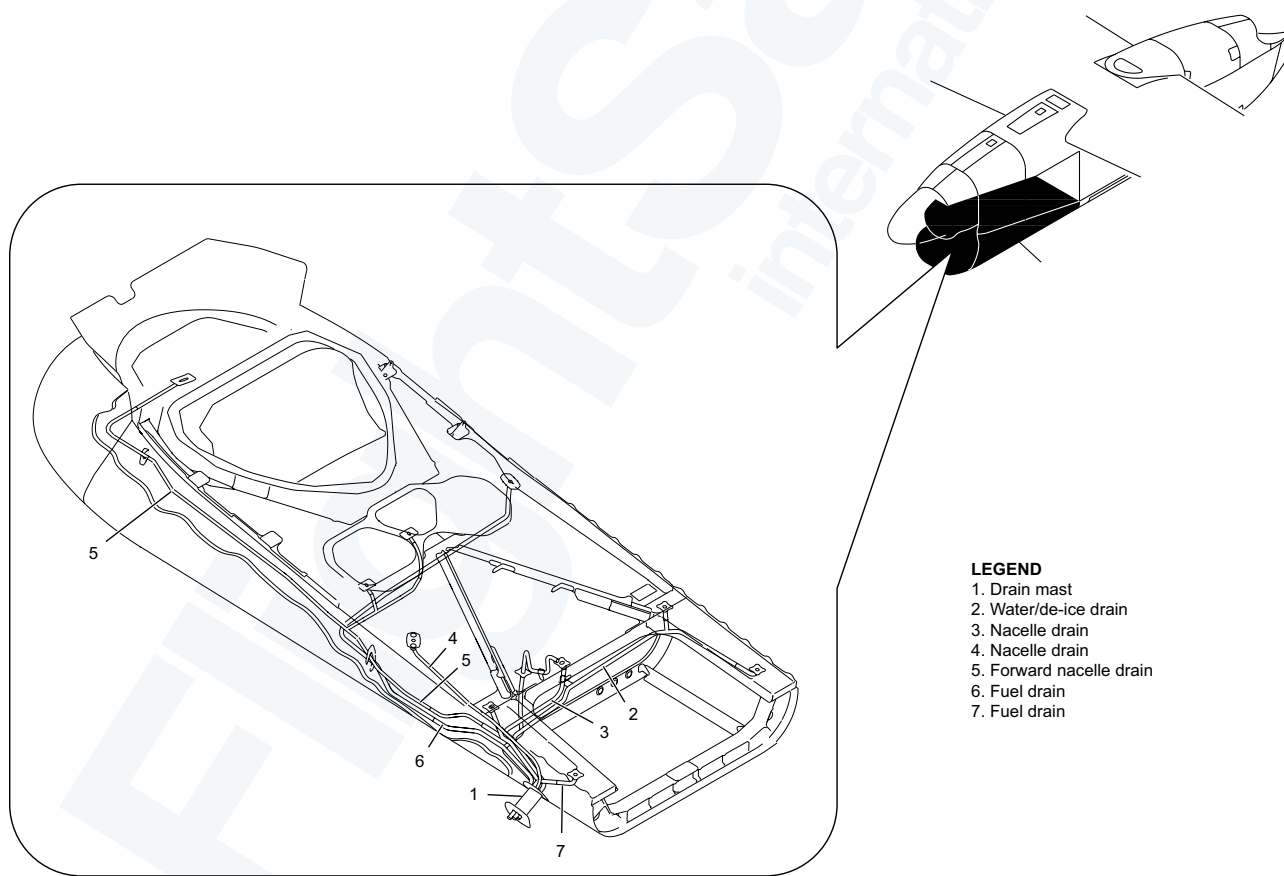


Figure 71-36. Forward Nacelle Air Intake Detail

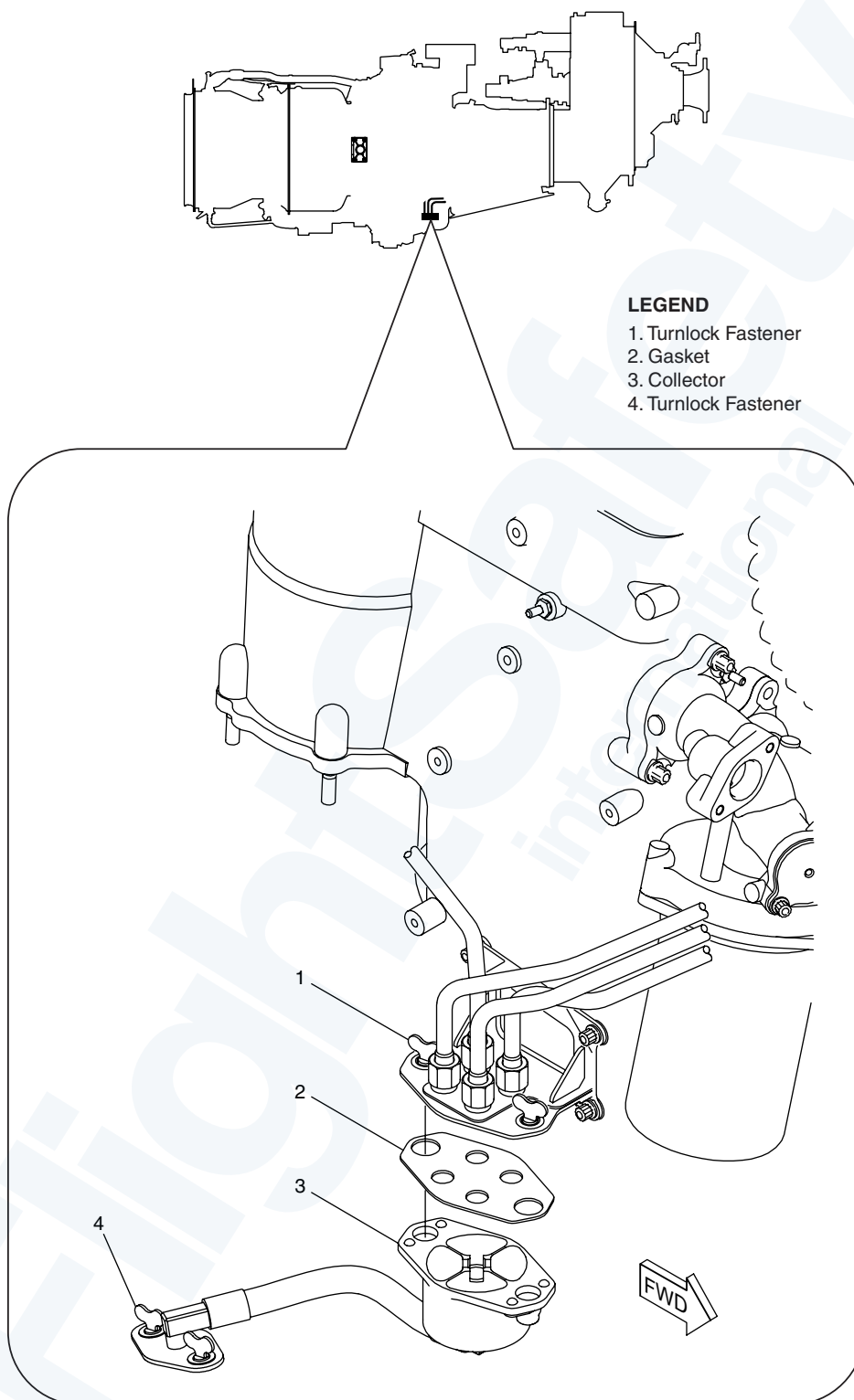
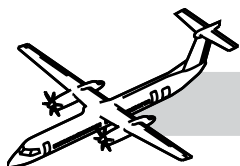
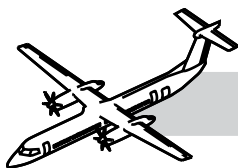


Figure 71-37. Forward Drain Collector



71-71-00 NACELLE DRAINS

INTRODUCTION

Engine nacelle drain lines safely dispose of fluid leakage.

GENERAL

The forward drain lines are routed to a collector tank. The rear drain lines are routed to a manifold.

SYSTEM DESCRIPTION

The forward and rear drains, the eight sumps and the Leading Edge Zone are routed separately to the aft end of the lower cowl and then overboard through an external drain mast.

The drain lines are comprised of the following sub-components:

- Fwd Drain Collector
- Rear Drain Manifold
- Drain mast
- Zone 1 - Engine Compartment / Fire Zone
- Zone 2 - Air intake Zone
- Zone 3 - Leading Edge Zone
- Tube assemblies.

COMPONENT DESCRIPTION

Forward Drain Collector

Refer to:

- Figure 71-37. Forward Drain Collector.
- Figure 71-38. Fwd Drain Collector.

The forward drain collector is on the right side of the engine aft of the scavenge oil filter.

The forward drain collector consists of a line removable drain collector with four labeled compartments.

The compartments are as follows:

- Outboard forward compartment - hydraulic and alternate feather pump drive seal drain
- Outboard aft compartment - PMA and fuel pump drive seal drain
- Inboard forward compartment - Propeller shaft seal drain
- Inboard aft compartment - Starter Generator drive seal drain.

The forward drain lines are routed individually to a collector tank with the exception of the drains for the auxiliary feather pump seal and engine driven hydraulic pump seal. Over flow of the collector tank connects a drain line to the lower cowl via a flexible hose with two camloc fasteners. This attachment allows for quick release during removal or lowering of the lower cowl to the maintenance position. From the lower cowl fluid drains overboard through an external drain mast.



Figure 71-38. Fwd Drain Collector

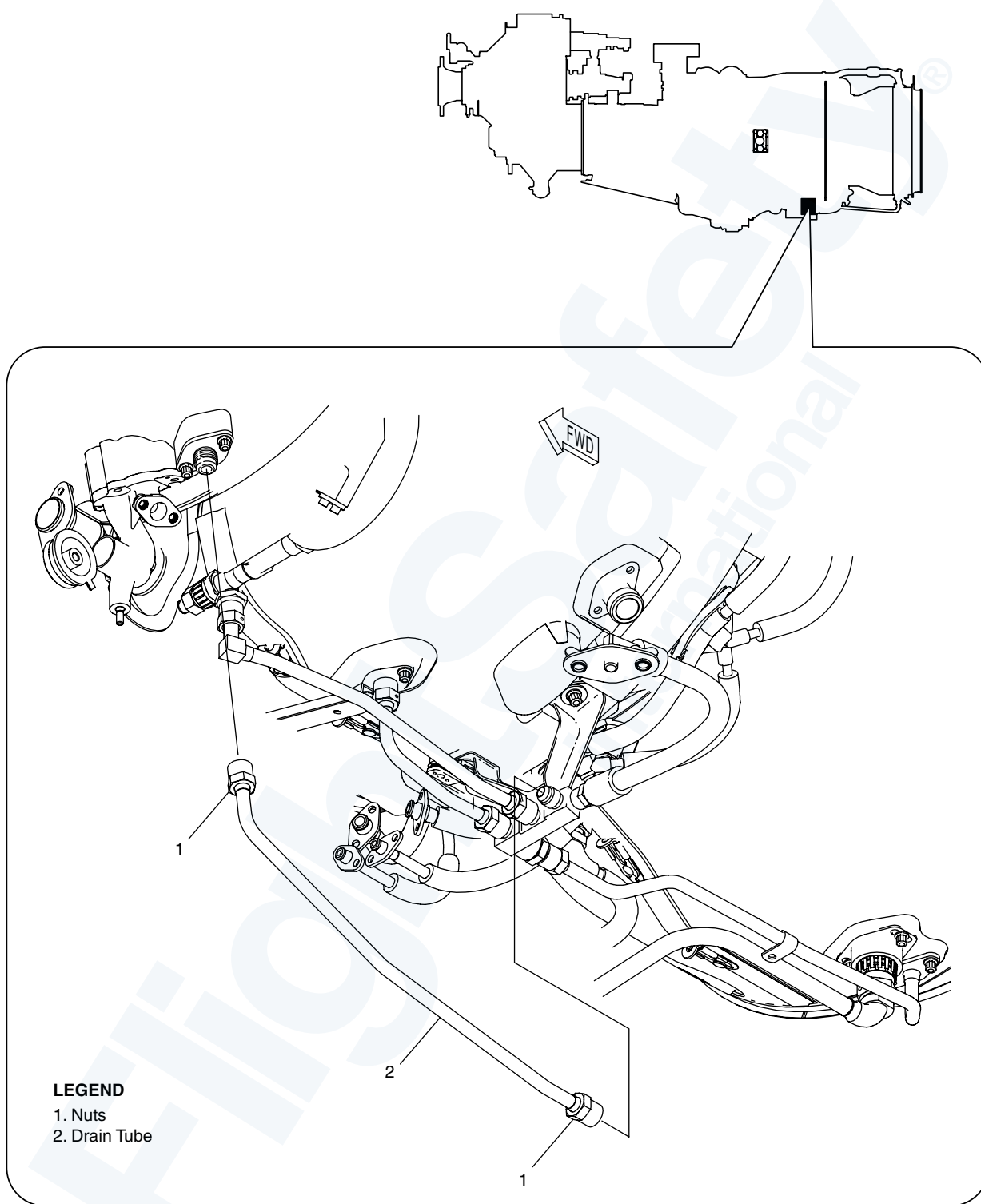
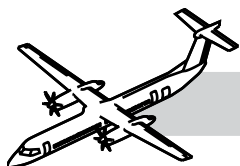
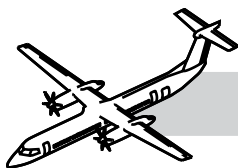


Figure 71-39. Rear Drain Manifold



Rear Drain Manifold

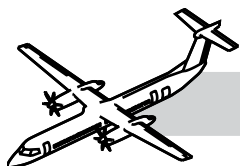
NOTE

Refer to Figure 71-37. Rear Drain Manifold.

The rear drain Manifold is on the lower left side of the engine aft of the oil sump.

The rear drain Manifold is a machined block with 5 fittings.

The rear drain lines are routed to a manifold which is connected to the lower cowl via a flexible hose with two camloc fasteners. From the lower cowl fluid drains overboard through an external drain mast. P3 air from the centrifuge provides the motive flow for this drain lines.



LEGEND

1. Drain mast
2. Water/de-ice drain
3. Nacelle drain
4. Nacelle drain
5. Forward nacelle drain
6. Fuel drain
7. Fuel drain

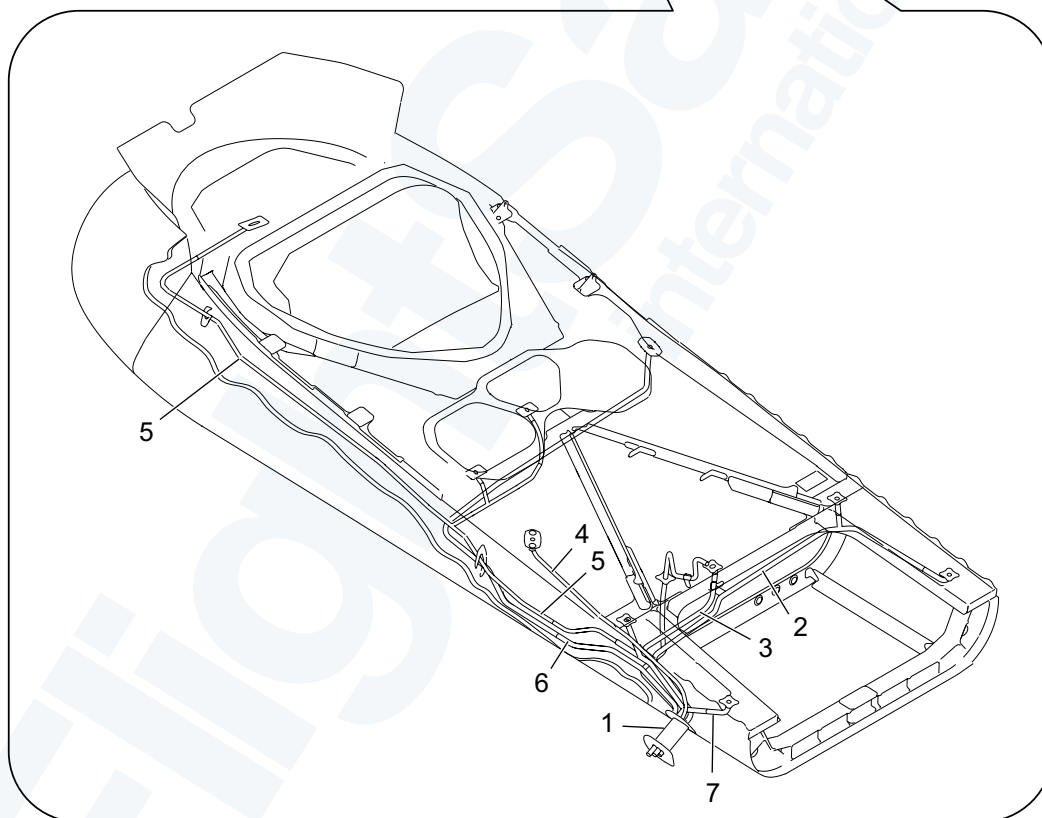
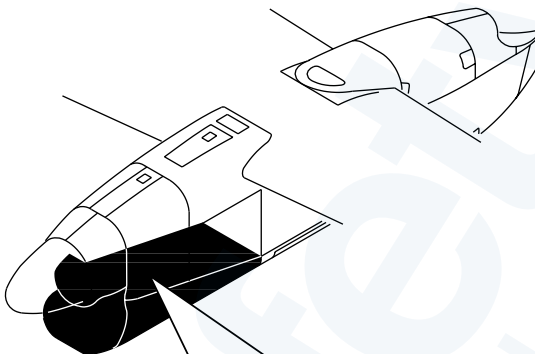
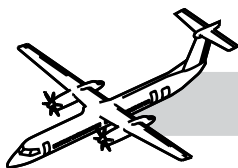


Figure 71-40. Forward Nacelle Air Intake Drains



Drain Mast

Refer to Figure 71-40. Forward Nacelle Air Intake Drains.

The drain mast on the left side of the lower cowl has drain ports for six tube assemblies listed below:

- Forward engine drain cup
- Rear engine drain Manifold
- Zone 1 sumps (three separate lines)
- Zone 3 fuel shroud drain.

The open ends of drain line assemblies in the mast are beveled to create a local negative pressure that enhance drainage into airflow away from the nacelle. The six tubes are arranged such that the efflux from each tube will not interfere with or be re-ingested by another tube.

There is also a run-back gate on the mast to prevent discharged fluids from running back along the outside of the mast and onto the nacelle surface. These drains have the capacity to discharge approximately 5 US gpm in flight, a flow rate comparable to an undetected fuel leakage rate.

Forward Nacelle Air Intakes Drains

Refer to Figure 71-40. Forward Nacelle Air Intake Drains.

Zone 1 - Engine Compartment/Fire Zone Eight sumps are on the firefloor of the lower cowl. Three drain holes are in the forward section and five are in the aft section. The sumps are low areas of the firefloor, connected to rigid tubing to the drain mast.

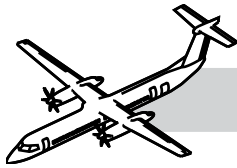
Fluids in the engine compartment can drain overboard through the eight sumps. The sumps form a part of the lower cowl drain system which merges and routes the drain lines to an external drain mast.

The Zone 2 - air intake zone - is on the underside of the lower cowl. There are four 0.27" diameter holes. Fluids in the lower cowl are drained overboard via drain holes in the bottom of the cowl.

The Zone 3 - Leading Edge Zone (LEZ) - is on the leading edge zone. Fluids in this zone are drained via a stainless steel tube.

Fluids accumulating in the forward part of the wing leading edge and de-icing system, drain into the right, rear drain tube and drain overboard.

The drain is on the port side of the nacelle at the lowest part of the zone, and is routed through the firezone and through the lower cowl to the external drain mast. The tubing runs vertically down the aft firewall and along the top of the oil cooler cover where it is connected to a line in the lower cowl. The fuel shroud drain is merged with this line just below the PEZ.



71-00-00 APPENDIX

MAINTENANCE CONSIDERATION

The following are abbreviated descriptions of the maintenance practices and are intended for training purposes only. For a more detailed description of the practices, refer to the tasks in the Bombardier AMM PSM 1-84-2.

Safety Precautions

WARNING

Torque the nacelle mounted hoist mounting bolts to the specified torque. This will help to prevent injuries to personnel, or damage to aircraft and equipment.

WARNING

Make sure that the hydraulic pump is correctly attached to the nacelle. If you do not do this, when you move the engine, it can cause injury to persons or damage to the pump or the engine.

CAUTION

Make sure that the drive shaft does not fall out of the pump when you disengage the pump from the engine. If the drive shaft falls, it can be damaged.

CAUTION

Make sure that the intake air passage is clear of unwanted objects before installation. If you do not do this, it can cause damage to the engine.

CAUTION

Do not lift the engine when it is attached to the stand. You can cause damage to the engine.

CAUTION

Keep all the parts together when you do work on the isolator. Do not mix the parts of different isolators. If you do, it is possible that you will not correctly monitor the structural life of the isolator parts. This can cause a dangerous flight condition to occur.

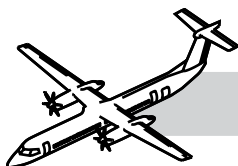
CAUTION

Do not use sharp objects or tools when you examine the bond separation on the isolator assemblies. Carefully use only finger pressure to examine the isolator assemblies. Sharp objects can cause damage to the isolators.

Special Tooling

The following special tools are used:

- GSB0700024
- Stand - Tail Support
- PWC55453
- Sling
- GSB7100022
- Hoist, Nacelle Mounted, Engine (replaces Hoist GSB7100021).



ENGINE OIL PRESSURE CHECK AND ADJUSTMENT

The maintenance procedure that follows is used to do an engine oil pressure check and make adjustments as necessary.

Engine Oil Pressure Check:

- Start the engine
- Operate the engine at 80% NH. When the oil temperature is 70° to 90°C (158°-194°F). Make sure that the oil pressure is 61 to 72 psid (421–496 kPa)
- Move the CLA to MAX/1020
- Move the PLA to RATING. Make sure that the oil pressure is 61 to 72 psid (421–496 kPa)
- Move the PLA to FI. Make sure the oil pressure is 61 to 72 psid (421–496 kPa)
- Move the PLA to DISC. If the oil pressure is less than 44 psid (304 kPa), replace the oil pressure regulating valve
- Shutdown the engine
- If the oil pressure is not in the limits, continue with the adjustment procedure.

Oil Pressure Adjustment:

- Remove the cover
- Remove and discard the packing
- Turn the adjuster clockwise to increase the oil pressure or counter clockwise to decrease the pressure. If it is necessary to go outside the limits of the adjuster to get the correct oil pressure, contact your local Field Support Representative.

CAUTION

WHEN YOU MAKE AN ADJUSTMENT TO THE OIL PRESSURE, THE FINAL POSITION OF THE ADJUSTER MUST BE BETWEEN 7 AND 11 TURNS COUNTER-CLOCKWISE FROM THE FULLY SEATED POSITION.

IF IT IS NECESSARY TO GO OUTSIDE THESE LIMITS TO GET THE CORRECT OIL PRESSURE, THEN EITHER THE PRESSURE REGULATING VALVE OR THE PRESSURE INDICATION IS DEFECTIVE.

NOTE:

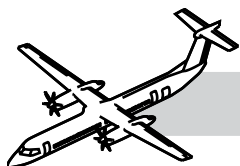
One complete turn of the adjuster changes the oil pressure by 3.5 to 4.0 psid (24.14-27.57 kPa).

Lubricate a new packing with engine oil (03-06) and install it on the cover.

Install the cover.

Torque the bolts to 36 to 40 lbf in (4.1–4.5 Nm).

Do the oil pressure check again to make sure that the oil pressure is correct.



ENGINE START

CAUTION

DO NOT TRY TO START THE ENGINE IF THE TEMPERATURE OF THE OIL IS BELOW -54°C (-65°F). YOU MUST HEAT THE OIL TO A TEMPERATURE ABOVE -54°C (-65°F). IF YOU DO NOT DO THIS, YOU CAN CAUSE DAMAGE TO THE ENGINE.

CAUTION

MAKE SURE THAT THE ENGINE INTAKE AIR PASSAGE IS CLEAR OF UNWANTED OBJECTS. IF YOU DO NOT DO THIS, IT CAN CAUSE DAMAGE TO THE ENGINE.

Install chocks on the main and nose landing gear wheels.

Engage the nose gear ground lock.

Install the main landing gear lock pins.

Set the Parking Brake to "ON."

NOTE:

When engine start is with No.1 engine, make sure the parking brake pressure reads 1000 psi minimum.

NOTE:

When engine start is with No.2 engine, make sure the parking brake pressure reads 500 psi minimum.

CAUTION

DISCONNECT THE EXTERNAL AC POWER SOURCE CURRENT TO THE AIRCRAFT. IF YOU DO NOT DO THIS, DAMAGE TO THE TRANSFORMER RECTIFIER UNIT COULD OCCUR.

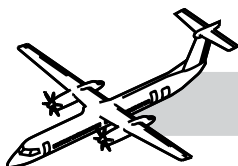
CAUTION

DO NOT OPERATE THE ENGINE ON THE GROUND IN CONTINUOUS CROSSWINDS OF MORE THAN 50 KNOTS OR IN GUSTING CROSSWINDS OF MORE THAN 55 KNOTS.

DO NOT OPERATE THE ENGINE ON THE GROUND AT ENGINE POWER ABOVE 460 SHP IN CROSSWINDS OF MORE THAN 45 KNOTS. IF YOU DO NOT OBEY THESE LIMITATIONS, YOU MUST REMOVE THE PROPELLER FROM SERVICE WITHIN THE NEXT 10 FLYING HOURS.

IMMEDIATELY DOWNLOAD THE FLIGHT DATA RECORDER AND SEND THE DATA TO BOMBARDIER AND DOWTY FOR EVALUATION. IF AVAILABLE, PLEASE PROVIDE BOMBARDIER AND DOWTY WITH THE AIRPORT REPORTED CONTINUOUS AND MAXIMUM GUST WIND CONDITIONS.

DO NOT USE MORE THAN 14% ENGINE TORQUE WITH UNDERSPEED GOVERNING AT 660 RPM. THIS WILL MAKE SURE THAT THE ENGINE POWER IS NOT MORE THAN 460 SHP.

**Pre-Start Checks:**

- IGNITION 1 and IGNITION 2 switches - NORM
- A/COL light switch - RED
- BLEED 1 switch - OFF
- BLEED 2 switch - OFF
- DOOR warning and FUELING ON caution lights - OFF
- PLA - DISC
- CLA - FUEL OFF.

Start one of the engines:

ENGINE START SELECT switch - 1 or 2.
Make sure that the SELECT light comes on.

CAUTION

DO NOT MOTOR THE ENGINE WITH THE STARTER FOR MORE THAN 70 SECONDS. THIS CAN CAUSE DAMAGE TO THE STARTER.

Engine START switch - Press. Make sure that the START light comes on. Abort the start if there is no NH indication and do the troubleshooting as necessary.

NOTE:

The starter is automatically de-energized when NH reaches 50%.

When you see an NH indication, continue with the steps that follow.

CAUTION

IF THE ENGINE DOES NOT LIGHT WITHIN 16 SECONDS OR IF NH DOES NOT GO TO 64.2% WITHIN 70 SECONDS AFTER YOU MOVE THE CONDITION LEVER TO START/FEATHER, DO THE STEPS THAT FOLLOW:

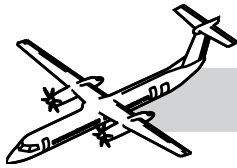
- MOVE THE CONDITION LEVER TO FUEL OFF
- ALLOW THE FUEL TO DRAIN FOR 30 SECONDS MINIMUM
- MOVE THE ENGINE START SELECT SWITCH TO OFF. MAKE SURE THAT THE SELECT AND START LIGHTS GO OFF
- DO A DRY MOTORING RUN FOR 15 SECONDS MINIMUM.

CAUTION

IF YOU STOP THE START PROCEDURE, LET THE ENGINE FULLY STOP BEFORE YOU TRY THE START PROCEDURE AGAIN.

CLA - START & FEATHER.**Do the checks that follow:**

- Make sure that the engine accelerates to more than 64.2% NH
- Make sure that the ITT/T6 does not go more than 920°C (1688°F)
- Make sure that the engine START and ENGINE START SELECT switches are OFF
- Make sure that the oil pressure (MOP) is more than 44 psi (304 kPa).



- Make sure that the warning and caution lights that follow are OFF:
 - ENG OIL PRESS
 - ENG FUEL PRESS
 - ENG HYD PUMP.

If you used the battery to start the engine, make sure that the DC GEN light is OFF.

Use the same procedure as before to start the other engine.

DC CONTROL EXT PWR switch - OFF (if applicable).

Make sure that the DC CONTROL EXT PWR is OFF.

Make sure that the caution lights that follow are OFF:

- No. 1 DC GEN
- No. 2 DC GEN
- MAIN BATTERY
- AUX BATTERY
- STBY BATTERY.

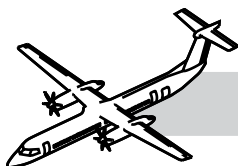
Disconnect the external power (if applicable).

MAIN BUS TIE switch - TIED.

ENGINE SHUTDOWN

Shutdown the engines:

- PARK/EMERG BRAKE lever - PARK
- Make sure that the STBY HYD PRESS ON and the PTU CNTL ON lights are OFF
- POWER levers - DISC
- Condition levers - START/FEATHER.
- Operate for 30 seconds minimum in feather
- STEERING switch - OFF
- ANTI SKID switch - OFF
- BLEED 1 and BLEED 2 switches - OFF
- RECIRC fan switch - OFF
- EMER LIGHTS switch - OFF
- Condition levers - FUEL OFF
- MAIN BATTERY, AUX BATTERY and STBY BATTERY switches - OFF
- BATTERY MASTER switch - OFF.



ENGINE DRY MOTORING

Do a dry motoring cycle of the engine as follows:

- Move the CLA to FUEL OFF
- Move the PLA to DISC
- Select IGNITION 1 or IGNITION 2 switch to the OFF position
- Move the ENGINE START SELECT switch to 1 or 2. Make sure that the SELECT light is ON.

CAUTION

DO NOT MOTOR THE ENGINE WITH THE STARTER FOR MORE THAN 70 SECONDS. THIS CAN CAUSE DAMAGE TO THE STARTER.

Push the Engine START switch. Motor the engine for 30 seconds.

Make sure that NH starts to increase. The duty cycles for the Engine DC starter are:

- Attempt No.1, 70 seconds cranking and 2 minutes recovery
- Attempt No.2, 70 seconds cranking and 2 minutes recovery
- Attempt No.3 70 seconds cranking and 30 minutes recovery
- Move the ENGINE SELECT switch to CENTER. Make sure that the SELECT and START lights go OFF.

ENGINE WET MOTORING

CAUTION

DISCONNECT THE EXTERNAL AC POWER SOURCE CURRENT TO THE AIRCRAFT. IF YOU DO NOT DO THIS, DAMAGE TO THE TRANSFORMER RECTIFIER UNIT COULD OCCUR.

CAUTION

ALWAYS DRY MOTOR THE ENGINE AFTER YOU WET MOTOR IT. THIS REMOVES FUEL FROM THE ENGINE.

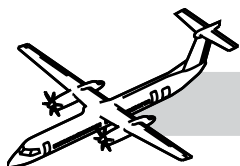
Do a wet motoring cycle of the engine as follows:

- CONDITION lever - FUEL OFF
- POWER lever - DISC
- IGNITION 1 or IGNITION 2 switch - OFF
- ENGINE START SELECT switch - 1 or 2. Make sure that the SELECT light is ON.

CAUTION

DO NOT MOTOR THE ENGINE WITH THE STARTER FOR MORE THAN 70 SECONDS. THIS CAN CAUSE DAMAGE TO THE STARTER.

- Engine START switch - Press. Move the CONDITION lever to START & FEATHER and motor the engine for 30 seconds
- CONDITION lever - FUEL OFF
- ENGINE START SELECT switch - OFF.



ENGINE POWER ASSURANCE CHECK

Do a manual power assurance check of the engine as follows:

- Record the outside air temperature (OAT) and the pressure altitude.

The correct pressure altitude may be obtained as follows:

- Set the altimeter calibration window to the standard barometric pressure (29.92 inch of mercury (in Hg) or 1013 Mbars). The altimeter reading is the actual pressure altitude around the aircraft. Do not use corrected barometric pressure as supplied by the tower
- Find and make a record of the maximum NH, NL, ITT and Wf.

CAUTION

MAKE SURE THAT THE ENGINE INTAKE AIR PASSAGE IS CLEAR OF UNWANTED OBJECTS. IF YOU DO NOT DO THIS, IT CAN CAUSE DAMAGE TO THE ENGINE.

- Start the engine
- Condition lever to MAX (1020 RPM)
- Advance the POWER lever to the rating detent. Make sure that the engine torque matches the target torque, the ECS bleed is OFF and airspeed is less than 65 Kts (calibrated). Operate the engine at this power for three minutes.
- Record NH, NL, ITT and Wf from the cockpit gages
- Shutdown the engine
- Make sure that OAT and the pressure altitude has not changed
- Using the tables determine the corrected margins

- Do the necessary troubleshooting if NH, NL, ITT and Wf margins are negative.

NOTE:

The engine is serviceable if Wf is more than the limit but all other parameters are in the limits. You can use Wf as an indicator. Make sure that the other parameters (NH, NL, ITT) are accurate.

- If ITT margin is less than 5°C (9°F) or the NH or NL margin is less than 0.25% schedule another PAC at an interval based on the engines observed performance deterioration rate, based on ECTM and previous PAC results.

Do a power assurance check of the engine using an EMU as follows:

- Select POWER ASSURANCE from the power plant main menu of the ARCDU
- Follow the ARCDU automated procedure.

Examine the engine parameters and calculated margins:

- Do the necessary troubleshooting if NH, NL, ITT or Wf margins are negative.

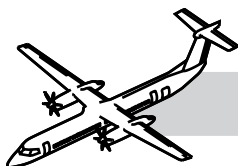
ENGINE ACCELERATION CHECK

Do an acceleration check of the engine as follows:

- Record the outside air temperature (T1.8) and the pressure altitude.

Do the steps that follow to find and record the target torque:

- Start the engine
- Move the CLA to MAX 1020
- Accelerate slowly until you reach the rating detent
- Maintain this power setting a few



seconds to let engine parameters stabilize

- Record RATING TORQUE
- Reduce power to flight idle
- Determine the torque value equivalent to 95% RATING TORQUE recorded in step 6, above.

CAUTION

PULL THE POWER LEVER BACK WHEN THE ENGINE GETS TO 95% TORQUE RATING. THIS WILL KEEP THE ENGINE OR THE PROPELLER IN THE PERMITTED LIMITS.

- Move the PLA from FLT IDLE to the RATING Detent position in less than one second to do a slam acceleration
- Record the amount of time it takes the engine to go from FLT IDLE to 95% of RATING TORQUE. The time taken must be 5 seconds maximum
- If the acceleration time is less than 5 seconds, the FMU is serviceable.

GROUND IDLE ENGINE RUN

Do a ground idle check as follows:

- Start the engine
- POWER lever to "DISC"
- CONDITION lever to FEATHER
- ECS bleed - OFF
- Check that NH is $64.2 \pm 0.5\%$.

**PROPELLER FORWARD
CONSTANT SPEED****CAUTION**

MAKE SURE THAT THE ENGINE INTAKE AIR PASSAGE IS CLEAR OF UNWANTED OBJECTS. IF YOU DO NOT DO THIS, IT CAN CAUSE DAMAGE TO THE ENGINE.

Do the check of the propeller forward constant speed as follows:

- Start the engine
- Move the CLA lever to MAX 1020.

CAUTION

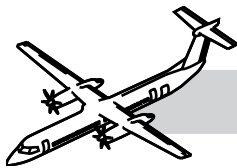
DO NOT OPERATE THE ENGINE AT MORE THAN THE TORQUE OR ITT LIMITS. IF YOU DO, YOU CAN CAUSE DAMAGE TO THE ENGINE OR THE PROPELLER.

- Advance the PLA until NP becomes stable (NP governing)
- Check that NP at MAX GOV is 1020 ± 10 rpm.

NOTE:

The NP gauge on the aircraft instrument panel moves in increments of 10 rpm.

- Move the PLA to DISC.



REVERSE MAXIMUM GOVERNING (FULL RANGE IN REVERSE)

Do the check of the reverse maximum governing as follows:

- Start the engine
- Move the CLA to any position between MIN 850 and MAX 1020
- Make sure the ECS bleed is OFF
- Move the PLA to MAX REV
- Check that NP becomes stable between 950 and 1030 rpm
- Move the PLA to DISC
- Move the CLA to START & FEATHER.

N_PT UNDERSPEED GOVERNING CHECK

Do a NPT underspeed governing check as follows:

- Start the engine
- Move the CLA to MAX/1020 and let the oil temperature become stable
- Move the PLA to FLIGHT IDLE
- Check that NP is minimum 660 ± 10 RPM.

FADEC TRIM (FOR PLA)

The PLA/CLA trim procedure does the following:

- Calibrates the PLA and CLA positions in the memory of the FADEC and PEC
- Uploads the Torque Trim values from the trim plug into the FADEC memory
 - Uploads the ITT Trim value from the trim resistor into the FADEC memory.

NOTE

If there is a difference between the actual trim resistor values and the values stored in the FADEC memory, the fault codes that follow will be displayed:

- Fault codes 796 and 797 for Torque Trim.
- Fault code 798 for ITT Trim.

If the PLA/CLA trim is required because of maintenance activity on the PLA or CLA quadrant, or because of an engine change, do the steps that follow:

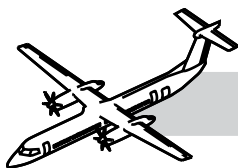
- Do an operational check for engine fault code indications
- If you see fault code 796, 797, 798, 804, 807 or 808, do the necessary maintenance before you do the PLA/CLA trim.

NOTE

The power settings or the ITT limit can be incorrect if you do the trim procedure before you clear these fault codes.

If the PLA/CLA trim is required because of an RGB change, do the steps that follow:

- Do an operational check for engine fault code indications
- If you see fault code 798 or 804, do the



necessary maintenance before you do the PLA/CLA trim.

NOTE

The power settings or the ITT limit can be incorrect if you do the trim procedure before you clear these fault codes.

If the PLA/CLA trim is required because of a FADEC change, and it had fault codes 797, 798, 804, 807 or 808 when it was removed, do the necessary maintenance before you do the PLA/CLA trim.

NOTE

The power settings or the ITT limit can be incorrect if you do the trim procedure before you clear these fault codes.

Do the steps that follow to trim the PLA and CLA:

- Move the PLA, of the powerplant to be trimmed, to FLIGHT IDLE and the CLA to 850 RPM
- Set the other PLA to the RATING position and the other CLA to FUEL OFF
- Set the MAINT DISC switch to ON
- Hold the RIG TRIM switch ON for more than 5 seconds.
- When you do the PLA Trim procedure, do the steps that follow:
 - To monitor the PLA Trim procedure, go to the PLA Trim page on the ARCDU display as follows:
 - Set the Central Maintenance System to Maintenance Mode).
 - At the ARCDU screen on the flight deck, select OTHER SYSTEMS from the main menu, then select EMU
 - From the main menu, select

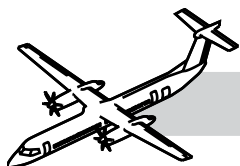
POWERPLANT INTERFACE , then select TRIM DATA

- At the TRIM DATA page, select PLA TRIM and read the values that come into view
- From the ARCDU screen, confirm that the trimmed PLA is 35° for both channels
- Move the two PLAs to FLIGHT IDLE
- Move the two CLAs to FUEL OFF
- On the Maintenance Control Panel, set the MAINT DISC switch to OFF.

PROPELLER FEATHER CHECK

Do a propeller feather check as follows:

- Start the engine
- Move the POWER lever to DISC
- Move the CONDITION lever to MAX/1020 and let the engine become stable. Make sure that the propeller fully unfeathers
- Move the CONDITION lever to START/FEATHER. Make sure that the propeller fully feathers within 20 seconds.



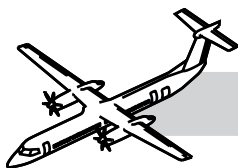
71-00-00 SPECIAL TOOLS & TEST EQUIPMENT

- GSB0700024 Stand - Tail Support
- GSB0700025 (or equivalent) Engine strap
- GSB7100021 Hoist, Nacelle Mounted, Engine
- GSB7100022 Hoist, Nacelle Mounted, Engine
- PWC55971 Stand, engine
- PWC55453 Sling
- PWC57111 Fixture, lifting
- GSB5411003 Cover - Engine Lower Cowl (Series 400)
- ACS28004 Sling - Lower Forward Cowl (Series 400)
- ACS28009 Hoist - Lower Forward Cowl (Series 400)
- ACS28005 Stand - Lower Forward Cowl Support/Transport (Series 400)
- GSB5400001 Bar - Pry
- ACS28016 Lanyard - Lower Forward Cowl (Series 400)
- Commercially available Load cell (able to measure a 2500 lbs (1134 kg) load)
- T900314-1 HTCS Ground Support Kit
- T900314-01 Hydraulic Pump Assembly (part of HTCS Ground Support Kit)
- T900314-03 Bleed Tube Assembly (part of HTCS Ground Support Kit)
- T900317-1 Alignment Tool (part of HTCS Ground Support Kit)

71-00-00 MAINTENANCE PRACTICES

Refer to the Bombardier AMM PSM 1-84-2 for details on these maintenance procedures:

- AMM 72-00-00-510-801: Shipping Methods.
- AMM 72-00-00-890-804: Engine Condition Trend Monitoring (ECTM).
- AMM 72-00-00-890-805: Oil Consumption Trend Monitoring.
- AMM 71-00-00-000-801: Removal of the Engine from the Nacelle.
- AMM 71-00-00-400-801: Installation of the Engine in the Nacelle.
- AMM 71-00-00-000-804: Removal of the Engine Hoist.
- AMM 71-00-00-400-804: Installation of the Engine Hoist.
- AMM 71-00-00-780-801: Engine Oil Pressure Check and Adjustment.
- AMM 71-00-00-868-806: Engine Power Assurance Check.
- AMM 71-00-00-868-807: Engine Acceleration Check.
- AMM 71-00-00-868-811: Ground Idle Engine Run.



- AMM 71-00-00-868-813: Propeller Forward Constant Speed.
- AMM 71-00-00-868-815: Oil Consumption Trend Monitoring Run.
- AMM 71-00-00-868-816: NPT Underspeed Governing Check.
- AMM 71-00-00-868-817: FADEC Trim (for PLA).
- AMM 71-00-00-868-818: Propeller Purge.
- AMM 71-00-00-868-819: Propeller Feather Check.
- AMM 71-10-16-010-801: Opening of the Lower Cowl.
- AMM 71-10-16-410-801: Closing of the Lower Cowl.
- AMM 71-10-16-210-801: General Visual Inspection of the Nacelle Air Intake Seal (MRB# 716100-201).
- AMM 71-10-16-000-801: Removal of the Lower Cowl.
- AMM 71-10-16-400-801: Installation of the Lower Cowl.
- AMM 71-21-01-000-801: Removal of the Forward Left Isolator.
- AMM 71-21-01-400-801: Installation of the Forward Left Isolator.
- AMM 71-21-11-000-801: Removal of the Forward Top Isolator.
- AMM 71-21-11-400-801: Installation of the Forward Top Isolator.
- AMM 71-21-16-000-801: Removal of the Aft Left Isolator.
- AMM 71-21-16-400-801: Installation of the Aft Left Isolator.
- AMM 71-21-21-000-801: Removal of the Aft Right Isolator.
- AMM 71-21-21-400-801: Installation of the Aft Right Isolator.
- AMM 71-21-00-840-805: Restoration (remolding) of the Molded Assemblies of the Aft Right Isolator.
- AMM 71-22-00-210-804: General Visual Inspection of the Hydraulic Torque Compensation System (MRB#712200-202).
- AMM 71-22-00-220-801: Inspection of the Hydraulic Torque Compensation System Reservoir Level Indicator.
- AMM 71-22-00-820-801: Rigging of the Hydraulic Torque Compensation System Cylinder Piston.
- AMM 71-22-00-600-801: Servicing of the Hydraulic Torque Compensation System.
- AMM 71-22-00-720-801: Functional Check of the Hydraulic Torque Compensation System Reservoir Level Indicator (MRB#712200-201).
- AMM 75-31-16-000-801: Removal of the P3 Air Separator.
- AMM 75-31-16-400-801: Installation of the P3 Air Separator.